

CHAPTER V

DEFLECTION OF TRANSVERSALLY LOADED BEAMS

31. Differential Equation of the Deflection Curve.—In

the design of a beam the engineer is usually interested not only in the stresses produced by the loads acting but also in the deflections produced by these loads. In many cases, furthermore, it is specified that the maximum deflection shall not exceed a certain small portion of the span.

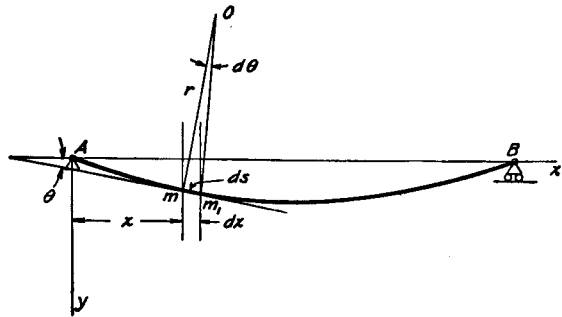


FIG. 114.

Let the curve AmB in Fig. 114 represent the shape of the axis of the beam after bending. This curve is called the *deflection curve*. To derive the differential equation of this curve we take the coordinate axes as shown in the figure and assume that the curvature of the deflection curve at any point depends only on the magnitude of the bending moment M at that point.¹ In such a case the relation between the curvature and the moment is the same as in the case of pure bending (see equation (56)), and we obtain

$$\frac{1}{r} = \frac{M}{EI_z} \quad (a)$$

¹ The effect of shearing force on the curvature will be discussed later (see art. 39). It will be shown that this effect is usually small and can be neglected.

To derive an expression for the relation between the curvature and the shape of the curve, we shall consider two adjacent points m and m_1 , ds apart on the deflection curve. If the angle which the tangent at m makes with the x axis is denoted by θ , the angle between the normals to the curve at m and m_1 is $d\theta$. The intersection point O of these normals gives the center of curvature and defines the length r of the radius of the curvature. Then

$$ds = r d\theta \quad \text{and} \quad \frac{1}{r} = \left| \frac{d\theta}{ds} \right|, \quad (b)$$

the bars indicating that we consider here only the numerical value of the curvature. Regarding the sign, it should be noted that the bending moment is taken positive in equation (a) if it produces upward concavity (see p. 71). Hence the curvature is positive when the center of curvature is above the curve as in Fig. 114. However, it is easy to see that for such a curvature the angle θ decreases as the point m moves along the curve from A to B . Hence, to a positive increment ds corresponds a negative $d\theta$. Thus to have the proper sign equation (b) must be written in the form

$$\frac{1}{r} = - \frac{d\theta}{ds}. \quad (c)$$

In practical applications only very small deflections of beams are allowable, and the deflection curves are very flat. In such cases we can assume with sufficient accuracy that

$$ds \approx dx \quad \text{and} \quad \theta \approx \tan \theta = dy/dx. \quad (d)$$

Substituting these approximate values for ds and θ in equation (c) we obtain

$$\frac{1}{r} = - \frac{d^2y}{dx^2}. \quad (e)$$

Equation (a) thus becomes

$$EI_z \frac{d^2y}{dx^2} = - M. \quad (79)$$

This is the differential equation of the deflection curve which must be integrated in each particular case to find deflections of beams.

It should be noted that the sign in equation (79) depends upon the direction of the coordinate axes. For example, if we take y positive upwards, it is necessary to put

$$\theta \approx - dy/dx$$

in place of equation (d); and we obtain plus instead of minus on the right side of equation (79).

In the case of very slender bars, in which the deflection may be large, it is not permissible to use the simplifications (d); and we must have recourse to the exact expression

$$\theta = \arctan \left(\frac{dy}{dx} \right).$$

Then

$$\begin{aligned} \frac{1}{r} &= -\frac{d\theta}{ds} = -\frac{d \arctan \left(\frac{dy}{dx} \right)}{dx} \frac{dx}{ds} \\ &= -\frac{\frac{d^2y}{dx^2}}{\left[1 + \left(\frac{dy}{dx} \right)^2 \right]^{3/2}}. \quad (f) \end{aligned}$$

Comparing this result with equation (e), it can be concluded that the simplifications shown in equation (d) are equivalent to assuming that the quantity $(dy/dx)^2$ in the denominator of the exact formula (f) is small in comparison with unity and can therefore be neglected.²

By differentiating equation (79) with respect to x and using equations (50) and (51), we obtain

$$EI_z \frac{d^3y}{dx^3} = -V$$

² The exact expression (f) for the curvature was used by the first investigators of the deflection curves. It was used, for example, by L. Euler in his famous work on "Elastic Curves," an English translation of which was published in "Isis," No. 58 (vol. XX, 1) November, 1933.

and

$$EI_z \frac{d^4y}{dx^4} = q. \quad (80)$$

The last equation is sometimes used in considering the deflection of beams under a distributed load.

32. Bending of a Uniformly Loaded Beam.—In the case of a *simply supported beam* which is uniformly loaded, Fig. 63, the bending moment at a cross section mn , a distance x from the left support, is

$$M = \frac{qlx}{2} - \frac{qx^2}{2}$$

and the differential equation (79) becomes

$$EI_z \frac{d^2y}{dx^2} = -\frac{qlx}{2} + \frac{qx^2}{2}.$$

Multiplying both sides by dx and integrating, we obtain

$$EI_z \frac{dy}{dx} = -\frac{qlx^2}{4} + \frac{qx^3}{6} + C \quad (a)$$

where C is the constant of integration which is to be adjusted to satisfy the conditions of this particular problem. To this end, we note that as a result of symmetry the slope at the middle of the span is zero. Setting $dy/dx = 0$ when $x = l/2$, we thus obtain

$$C = \frac{ql^3}{24}$$

and equation (a) becomes

$$EI_z \frac{dy}{dx} = -\frac{qlx^2}{4} + \frac{qx^3}{6} + \frac{ql^3}{24}. \quad (b)$$

A second integration gives

$$EI_z y = -\frac{qlx^3}{12} + \frac{qx^4}{24} + \frac{ql^3x}{24} + C_1. \quad (c)$$

The new constant of integration C_1 is determined from the condition that the deflection at the supports is zero. Sub-

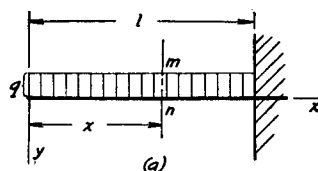
stituting $y = 0$ and $x = 0$ into equation (c) we find $C_1 = 0$. Equation (c) then becomes

$$y = \frac{q}{24EI_z} (l^3x - 2lx^3 + x^4). \quad (81)$$

This is the deflection curve of a simply supported and uniformly loaded beam. The maximum deflection of this beam is evidently at the middle of the span. Substituting $x = l/2$ in equation (81) we thus find

$$y_{\max} = \frac{5}{384} \frac{ql^4}{EI_z}. \quad (82)$$

The maximum slope occurs at the left end of the beam where, by substituting $x = 0$ in equation (b), we obtain



$$\left(\frac{dy}{dx} \right)_{\max} = \frac{ql^3}{24EI_z}. \quad (83)$$

In the case of a *uniformly loaded cantilever beam*, Fig. 115a, the bending moment at a cross section mn a distance x from the left end is

$$M = -\frac{qx^2}{2},$$

and equation (79) becomes

$$EI_z \frac{d^2y}{dx^2} = \frac{qx^2}{2}.$$

The first integration gives

$$EI_z \frac{dy}{dx} = \frac{qx^3}{6} + C. \quad (a)$$

The constant of integration is found from the condition that the slope at the built-in end is zero, that is $dy/dx = 0$ for $x = l$. Substituting these values in equation (a) we find

$$C = -\frac{ql^3}{6}.$$

The second integration gives

$$EI_z y = \frac{qx^4}{24} - \frac{ql^3x}{6} + C_1. \quad (b)$$

The constant C_1 is found from the condition that the deflection vanishes at the built-in end. Thus, by substituting $x = l$, $y = 0$ in equation (b), we obtain

$$C_1 = \frac{ql^4}{8}.$$

Substituting this value in equation (b), we find

$$y = \frac{q}{24EI_z} (x^4 - 4l^3x + 3l^4). \quad (84)$$

This equation defines the deflection curve of the uniformly loaded cantilever.

If the left end, instead of the right end, is built in, as in Fig. 115b, the deflection curve is evidently obtained by substituting $l - x$ instead of x in equation (84). In this way we find

$$y = \frac{q}{24EI_z} (x^4 - 4lx^3 + 6l^2x^2). \quad (85)$$

Problems

1. A simply supported and uniformly loaded wooden beam of square cross section has a span $l = 10$ ft. Find the maximum deflection if $(\sigma_x)_{\max} = 1,000$ lbs. per sq. in., $E = 1.5 \times 10^6$ lbs. per sq. in. and $q = 400$ lbs. per ft.

2. Find the depth of a uniformly loaded and simply supported steel I beam having a span of 10 ft., if the maximum bending stress is 16,000 lbs. per sq. in. and the maximum deflection $\delta = 0.1$ in.

Answer: $h = 16$ in.

3. A uniformly loaded cantilever beam of a span $l = 10$ ft. has a deflection at the end equal to $0.01l$. What is the slope of the deflection curve at the end?

4. What is the length of a uniformly loaded cantilever beam if its deflection at the free end is 1 in. and the slope of the deflection curve at the same point is 0.01 ?

5. A uniformly loaded steel I beam supported at the ends has a deflection at the middle of $\delta = 5/16$ in. while the slope of the deflection curve at the end $\theta = 0.01$. Find the depth h of the beam if the maximum bending stress is $\sigma = 18,000$ lbs. per sq. in.

Solution:

We use the known formulas

$$\delta = \frac{5}{384} \frac{ql^4}{EI}, \quad \theta = \frac{ql^3}{24EI}, \quad \sigma_{\max} = \frac{ql^2}{8} \times \frac{h}{2I}.$$

From the first two formulas we find

$$\frac{5}{16} l = \frac{\delta}{\theta} = \frac{5}{16} \times 100 \text{ in.} \quad \text{and} \quad l = 100 \text{ in.}$$

The second formula then gives

$$\frac{ql^2}{8I} = \frac{3E\theta}{l} = \frac{3 \times 30 \times 10^6 \times 0.01}{100}.$$

Substituting this in the third formula, we obtain

$$h = \frac{2 \times 18,000 \times 100}{3 \times 30 \times 10^6 \times 0.01} = 4 \text{ in.}$$

33. Deflection of a Simply Supported Beam Loaded with a Concentrated Load.—In this case there are two different

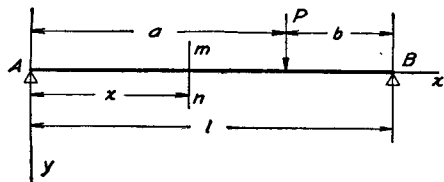


FIG. 116.

expressions for the bending moment (see p. 75) corresponding to the two portions of the beam, Fig. 116. Equation (79) for the deflection curve must therefore be written for

$$EI_z \frac{d^2y}{dx^2} = -\frac{Pb}{l} x \quad \text{for} \quad x \leq a$$

and

$$EI_z \frac{d^2y}{dx^2} = -\frac{Pb}{l} x + P(x - a) \quad \text{for} \quad x \geq a.$$

By integrating these equations we obtain

$$EI_z \frac{dy}{dx} = -\frac{Pbx^2}{2l} + C \quad \text{for} \quad x \leq a \quad \left. \vphantom{EI_z \frac{dy}{dx}} \right\} (a)$$

and

$$EI_z \frac{dy}{dx} = -\frac{Pbx^2}{2l} + \frac{P(x-a)^2}{2} + C_1 \quad \text{for} \quad x \geq a.$$

Since the two branches of the deflection curve must have a common tangent at the point of application of the load P , the above expressions (a) for the slope must be equal for $x = a$. From this we conclude that the constants of integration are equal, i.e., $C = C_1$. Performing the second integration and substituting C for C_1 , we obtain

$$EI_z y = -\frac{Pbx^3}{6l} + Cx + C_2 \quad \text{for} \quad x \leq a \quad \left. \vphantom{EI_z y} \right\} (b)$$

and

$$EI_z y = -\frac{Pbx^3}{6l} + \frac{P(x-a)^3}{6} + Cx + C_3 \quad \text{for} \quad x \geq a.$$

Since the two branches of the deflection curve have a common deflection at the point of application of the load, the two expressions (b) must be identical for $x = a$. From this it follows that $C_2 = C_3$. Finally we need to determine only two constants C and C_2 , for which determination we have two conditions, namely that the deflection at each of the two ends of the beam is zero. Substituting $x = 0$ and $y = 0$ in the first of expressions (b), we find

$$C_2 = C_3 = 0. \quad (c)$$

Substituting $y = 0$ and $x = l$ in the second of expressions (b) we obtain

$$C = \frac{Pbl}{6} - \frac{Pb^3}{6l} = \frac{Pb(l^2 - b^2)}{6l}. \quad (d)$$

Substituting the values (c) and (d) of the constants into equations (b) for the deflection curve, we obtain

$$EI_z y = \frac{Pbx}{6l} (l^2 - b^2 - x^2) \quad \text{for} \quad x \leq a \quad (86)$$

and

$$EI_z y = \frac{Pbx}{6l} (l^2 - b^2 - x^2) + \frac{P(x-a)^3}{6} \quad \text{for} \quad x \geq a. \quad (87)$$

The first of these equations gives the deflections for the left portion of the beam and the second gives the deflections for the right portion.

Substituting the value (d) into equations (a) we obtain

$$\left. \begin{aligned} EI_z \frac{dy}{dx} &= \frac{Pb}{6l} (l^2 - b^2 - 3x^2) \quad \text{for} \quad x \leq a \\ EI_z \frac{dy}{dx} &= \frac{Pb}{6l} (l^2 - b^2 - 3x^2) + \frac{P(x-a)^2}{2} \quad \text{for} \quad x \geq a. \end{aligned} \right\} \quad (e)$$

From these equations the slope at any point of the deflection curve can readily be calculated. Often we need the values of the slopes at the ends of the beam. Substituting $x = 0$ in the first of equations (e), $x = l$ in the second, and denoting the slopes at the corresponding ends by θ_1 and θ_2 we obtain³

$$\theta_1 = \left(\frac{dy}{dx} \right)_{x=0} = \frac{Pb(l^2 - b^2)}{6lEI_z}, \quad (88)$$

$$\theta_2 = \left(\frac{dy}{dx} \right)_{x=l} = -\frac{Pab(l+a)}{6lEI_z}. \quad (89)$$

The maximum deflection occurs at the point where the tangent to the deflection curve is horizontal. If $a > b$ as in Fig. 116, the maximum deflection is evidently in the left portion of the beam. We can find the position of this point by equating the first of the expressions (e) to zero to obtain

$$l^2 - b^2 - 3x^2 = 0,$$

³ For flat curves, which we have in most cases, the slopes θ_1 and θ_2 can be taken numerically equal to the angles of rotation of the ends of the beam during bending, the slopes being taken positive when the rotation is clockwise.

from which

$$x = \frac{\sqrt{l^2 - b^2}}{\sqrt{3}}. \quad (f)$$

This is the distance from the left support to the point of maximum deflection. To find the maximum deflection itself we substitute expression (f) in equation (86), which gives

$$y_{\max} = \frac{Pb(l^2 - b^2)^{3/2}}{9\sqrt{3}lEI_z}. \quad (g)$$

If the load P is applied at the middle of the span the maximum deflection is evidently at the middle also. Its magnitude is obtained by substituting $b = l/2$ in equation (g), which gives

$$(y)_{x=l/2} = \frac{Pl^3}{48EI_z}. \quad (90)$$

From equation (f) it can be concluded that in the case of one concentrated force the maximum deflection is always near the middle of the beam. When $b = l/2$ it is at the middle; in the limiting case, when b is very small and P is at the support, the distance x as given by equation (f) is $l/\sqrt{3}$, and the point of maximum deflection is only a distance

$$\frac{l}{\sqrt{3}} - \frac{l}{2} = 0.077l$$

from the middle. Due to this fact the deflection at the middle is a close approximation to the maximum deflection. To obtain the deflection at the middle we substitute $x = l/2$ in equation (86), which gives

$$(y)_{x=l/2} = \frac{Pb}{48EI_z} (3l^2 - 4b^2). \quad (91)$$

The difference of the deflections (g) and (91) in the most unfavorable case, that is when b approaches zero, is only about 2.5 per cent of the maximum deflection.

Problems

1. Find the position of the load P , Fig. 116, if the ratio of the numerical values of the slopes at the ends of the beam is $|\theta_1/\theta_2| = \frac{3}{4}$.
2. Find the difference between the maximum deflection and the deflection at the middle of the beam in Fig. 116 if $b = 2a$.
3. Find the maximum deflection of the beam shown in Fig. 116 if AB is an American Standard I beam, 8 in. in depth and 5.34 sq. in. in cross-sectional area, and $a = 12$ ft., $b = 8$ ft., and $P = 2000$ lbs.
4. What will be the maximum deflection if the I beam of the previous problem is replaced by a wooden beam having a cross section 10 in. by 10 in. The modulus of elasticity for wood can be taken as $E = 1.5 \times 10^6$ lbs. per sq. in.

34. Determination of Deflections by the use of the Bending Moment Diagram; Method of Superposition.—

In the preceding articles it was shown how the deflection curve of a beam can be determined by integration of the differential equation (79). In many cases, however, especially if we need the deflection in a prescribed point rather than the general equation of the deflection curve, the calculation can be considerably simplified by the use of the bending moment diagram as will be described in the following discussion.⁴

In Fig. 117 AB represents a portion of a deflection curve and a_1b_1 the corresponding portion of the bending moment

⁴ The use of the bending moment diagram in calculating deflections of beams was developed by O. Mohr, see Zeitschr. d. Architekten und Ingenieur-Vereins zu Hannover, p. 10, 1868. See also O. Mohr, Abhandlungen aus dem Gebiete der Technischen Mechanik, p. 294, Berlin, 1906. A similar method was developed independently of O. Mohr by Prof. C. E. Green, University of Michigan, 1874.

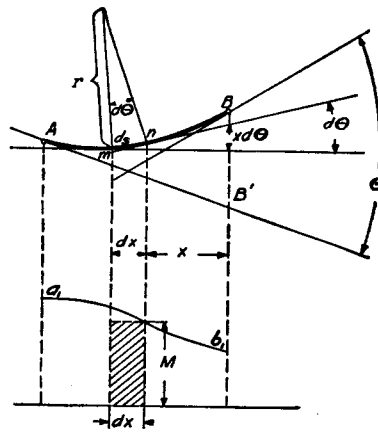


FIG. 117.

diagram. Two adjacent cross sections of the beam at distance ds apart will intersect after bending, at an angle $d\theta$, and, from eq. (56),

$$d\theta = \frac{1}{r} ds = \frac{M}{EI_z} ds.$$

For beams used in structures, the curvature is very small, and we may use dx for ds . Then

$$d\theta = \frac{1}{EI_z} (Mdx). \quad (a)$$

Graphically interpreted, this means that the elemental angle $d\theta$ between two consecutive radii or two consecutive tangents to the deflection curve equals the shaded elemental area Mdx of the bending moment diagram, divided by the flexural rigidity.⁵ This being so for each element, the angle θ between the tangents at A and B will be obtained by summarizing such elements as given by eq. (a). Then

$$\theta = \int_A^B \frac{1}{EI_z} Mdx, \quad (92)$$

that is, the angle between the tangents at two points A and B of the deflection curve equals the area of the bending moment diagram between the corresponding verticals, divided by the flexural rigidity of the beam.

Let us consider now the distance of the point B from the tangent AB' at point A . Recalling that a deflection curve is a flat curve, the above distance can be measured along the vertical BB' . The contribution made to this distance by the bending of an element mn of the beam and included between the two consecutive tangents at m and n is equal to

$$xd\theta = x \frac{Mdx}{EI_z}.$$

Interpreted graphically this is $1/EI_z$ (moment of shaded area Mdx with respect to the vertical through B). Integration

⁵ By way of dimensional check: $d\theta$ is in radians, i.e., a pure number, Mdx in inch lbs. $\times$ inches, EI_z in lbs. per sq. in. $\times$ (inches)⁴.

gives the total deflection BB' :

$$\overline{BB'} = \delta = \int_A^B \frac{1}{EI_z} x M dx, \quad (93)$$

that is, the distance of B from the tangent at A is equal to the moment with respect to the vertical through B of the area of the bending moment diagram between A and B , divided by the flexural rigidity EI_z . By using eqs. (92) and (93) the slope of the deflection curve and the magnitude of deflection at any cross section of the beam can easily be calculated. This method of calculating deflections is called *Area-Moment Method*.

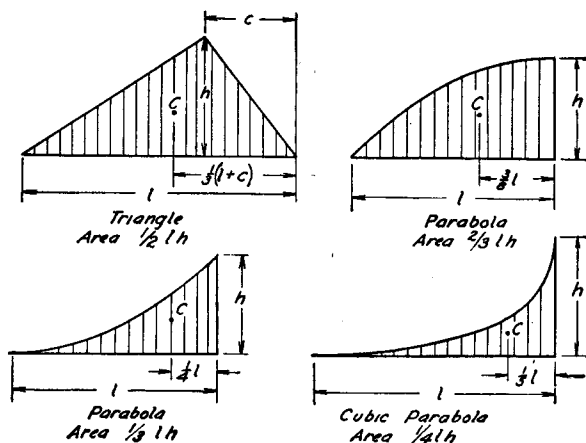


FIG. 118.

It should be noted that the deflection of a beam of a given flexural rigidity (see equation (93)) is entirely defined by the bending moment diagram. From this fact a very important conclusion can be drawn. It will be appreciated from the definition of the bending moment (art. 19) that the bending moment produced at a cross section mn of a beam by several simultaneously acting loads is equal to the sum of the moments produced at the same cross section by the individual loads acting separately. On the basis of this conclusion, together with equation (93), it can be stated that the deflection produced at a point of a beam by a system of simultane-

ously acting loads can be obtained by summing up the deflections produced at that point by the individual loads. For example, if the deflection curve produced by a single load (equations (86) and (87)) is known, the deflection produced by several loads is obtained by simple summation. This method of calculating deflections will be called *the method of superposition* in the subsequent discussion.

The calculation of the integrals in equations (92) and (93) can often be simplified by the use of known formulas concerning areas and centroids. Several formulas which are often encountered in applications are given in Fig. 118.

35. Deflection of a Cantilever Beam by the Area-Moment Method.—For the case of a cantilever beam with a concentrated load at the end (Fig. 119, *a*) the bending moment diagram is shown in Fig. 119 (*b*). Since a tangent at the built-in end A remains fixed, the distances of points of the deflection curve from this tangent are actual deflections. The angle θ_b which the tangent to the deflection curve at B makes with the tangent at A is called the *angular deflection* of B with respect to A . Then from eq. (92) ⁶

$$\theta_b = Pl \times \frac{l}{2} \times \frac{1}{EI_z} = \frac{Pl^2}{2EI_z}. \quad (94)$$

The deflection δ is calculated from eq. (93) as the moment of the area aba_1 about the vertical through b divided by EI_z . Then

$$\delta = Pl \times \frac{l}{2} \times \frac{2}{3} l \times \frac{1}{EI_z} = \frac{Pl^3}{3EI_z}. \quad (95)$$

For any cross section such as mn , the angular deflection from

⁶ The numerical value of the angular deflection is calculated. The direction of the deflection is readily seen from the loading conditions.

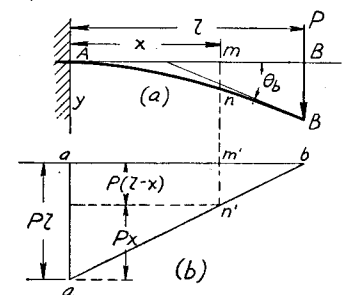


FIG. 119.

A is the area $m'n'aa_1$ of Fig. 119 (*b*), divided by EI_z . In the case of flat curves such as deflection curves of beams, angular deflection can be taken equal to the slope of the curve and we obtain

$$\theta = \frac{dy}{dx} = \frac{Pl^2}{2EI_z} \left[1 - \frac{(l-x)^2}{l^2} \right]. \quad (96)$$

The deflection y at the same cross section is the moment of the area $m'n'aa_1$ about $m'n'$ divided by EI_z (see eq. 93). Separating this area into the rectangle and the triangle indicated in the figure, this is

$$y = \frac{1}{EI_z} \left[P(l-x) \frac{x^2}{2} + \frac{Px^2}{2} \frac{2x}{3} \right] = \frac{P}{EI_z} \left(\frac{lx^2}{2} - \frac{x^3}{6} \right). \quad (97)$$

For a cantilever with a concentrated load P at a cross section a distance c from the support (Fig. 120, *a*) the bending moment diagram is shown in Fig. 120 (*b*). The slope and the deflection for any section to the left of the point of application of the load are determined from eqs. (96) and (97) with c in place of l . For any cross section to the right of the load the bending moment and the curvature are zero; hence this portion of the beam remains straight. The slope is constant and equal to the slope at D , i.e., from eq. (94), $Pc^2/2EI_z$. The deflection at any cross section mn is the moment of the area of the triangle aa_1d about the vertical $m'n'$ divided by EI_z , which gives

$$y = \frac{1}{EI_z} \frac{Pc^2}{2} \left(x - \frac{1}{3}c \right). \quad (98)$$

In the case of a cantilever with a uniform load of intensity q (Fig. 121, *a*) the bending moment at any cross section

mn distant x_1 from the built-in end is

$$M = -\frac{q(l-x_1)^2}{2}.$$

The slope at any cross section a distance x from the support is, from eq. (92),

$$\begin{aligned} \theta = \frac{dy}{dx} &= \frac{1}{EI_z} \int_0^x \frac{q(l-x_1)^2}{2} dx_1 \\ &= \frac{q}{2EI_z} \left(l^2x - lx^2 + \frac{x^3}{3} \right). \end{aligned} \quad (99)$$

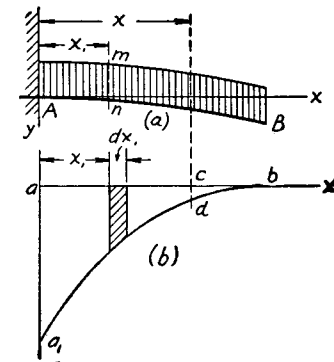


FIG. 121.

The slope at the end is obtained by substituting l for x in the above equation, giving

$$\left(\frac{dy}{dx} \right)_{x=l} = \frac{ql^3}{6EI_z}. \quad (100)$$

The deflection at any section a distance x from the built-in end is the moment of the area aa_1cd about the vertical cd divided by EI_z (Fig. 121, *b*). The moment of the element of this area, shown shaded, is

$$(x-x_1) \frac{q(l-x_1)^2}{2} dx_1$$

and the total moment is the integral of this with respect to x_1 from $x_1 = 0$ to $x_1 = x$. Hence

$$y = \frac{1}{EI_z} \frac{q}{2} \int_0^x (x-x_1)(l-x_1)^2 dx_1.$$

The deflection at any point a distance x from the support is then, after integration,

$$y = \frac{q}{2EI_z} \left(\frac{l^2x^2}{2} - \frac{lx^3}{3} + \frac{x^4}{12} \right). \quad (101)$$

For the deflection at the end $x = l$,

$$\delta = (y)_{x=l} = \frac{ql^4}{8EI_z}. \quad (102)$$

The same problem can be solved by using the method of superposition. The uniform load can be considered as a system of infinitesimal loads qdc as indicated by the shaded area in Fig. 122. The deflection produced at the cross section mn by each elemental load qdc to its left can be found from eq. (98) by substituting qdc for P . The deflection y_1 produced by the total load to the left of mn is the summation of the deflections produced

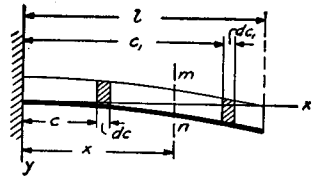


FIG. 122

by all such elemental loads with c varying from $c = 0$ to $c = x$:

$$y_1 = \frac{1}{EI_z} \int_0^x \frac{qc^2}{2} (x - \frac{1}{3}c) dc = \frac{q}{2EI_z} \frac{x^4}{4}.$$

The deflection produced at the cross section mn by an elemental load qdc_1 to its right is found from eq. (97) by substituting qdc_1 for P and c_1 for l . The deflection y_2 produced at mn by the total load to the right is the summation of the deflections due to all such elemental load, with c_1 varying from $c_1 = x$ to $c_1 = l$:

$$y_2 = \frac{1}{EI_z} \int_x^l q \left(\frac{c_1^3}{6} - \frac{x^2 c_1}{2} + \frac{x^3}{3} \right) dc_1 = \frac{q}{2EI_z} \left(-\frac{x^4}{6} + \frac{x^2 l^2}{2} - \frac{lx^3}{3} \right).$$

Then the total deflection at the section mn is

$$y = y_1 + y_2 = \frac{q}{2EI_z} \left(\frac{l^2 x^2}{2} - \frac{lx^3}{3} + \frac{x^4}{12} \right),$$

which agrees with eq. (101) found above.

Problems

1. Determine the deflection and the slope of the cantilever beam in problem 9, p. 108.

Solution.

$$\delta = \frac{Pl^3}{3EI_z} + \frac{ql^4}{8EI_z}.$$

2. Determine the deflection of the top of the pillar represented in Fig. 94.

Solution. The bending moment at any cross section mn , a distance x from the top, is

$$M = -\frac{Wx^3}{3l^2},$$

where $W = \frac{1}{2}dl^2 \times 62.4$ lbs. is the total hydrostatic pressure transmitted to one pillar. Using eq. (93), the deflection of the top of the pillar is

$$\delta = \frac{W}{EI_z} \int_0^l \frac{x^4 dx}{3l^2} = \frac{Wl^3}{15EI_z} = \frac{3 \times 6^2 \times 62.4 \times 6^3 \times 12^3 \times 12}{2 \times 15 \times 1.5 \times 10^6 \times 9.9^4} = 0.070 \text{ in.}$$

3. Determine the deflection and the slope at the end of a cantilever bent by a couple M (Fig. 123).

Answer.

$$(y)_{x=l} = -\frac{Ml^2}{2EI_z}; \quad \left(\frac{dy}{dx} \right)_{x=l} = -\frac{Ml}{EI_z}.$$

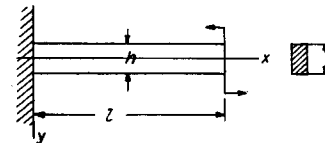


FIG. 123.

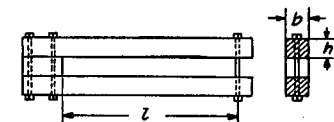


FIG. 124.

4. Two wooden rectangular beams clamped at the left end (Fig. 124) are bent by tightening the bolt at the right end. Determine the diameter d of the bolt to make the factors of safety for the wooden beams and for the steel bolt the same. The length of the beams $l = 3$ feet, the depth $h = 8$ in., the width $b = 6$ in., working stress for steel $\sigma_w = 12,000$ lbs. per sq. in., for wood $\sigma_w = 1,200$ lbs. per sq. in. Determine the deflection of the beams when the tensile stress in the bolt is 12,000 lbs. per sq. in.

Solution. If P is the force in the bolt, the equation for determining the diameter d will be

$$\frac{4P}{\pi d^2} : \frac{6Pl}{bh^2} = \frac{12,000}{1,200} = 10,$$

from which

$$d = 0.476 \text{ in.} \quad \text{and} \quad P = 12,000 \times \frac{\pi d^2}{4} = 2,130 \text{ lbs.}$$

Then from eq. (95), by taking $E = 1.5 \times 10^6$ lbs. per sq. in. the deflection $\delta = 0.0864$ in.

5. What is the ratio of the deflections at the ends of the cantilevers shown in Fig. 125 if the intensity of uniform load is the same in both cases?

Answer. 7 : 41.

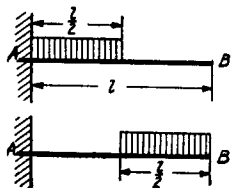


FIG. 125.

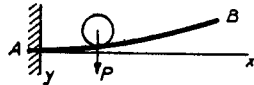


FIG. 126.

6. What must be the equation of the axis of the curved bar AB before it is bent if the load P , moving along the bar, remains always on the same level (Fig. 126)?

Answer.

$$y = -\frac{Px^3}{3EI_z}$$

7. Determine the safe deflection of the beam shown in Fig. 123 when the working stress σ_w is given. Determine this also for a cantilever loaded at the end (Fig. 119).

Answer.

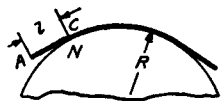


FIG. 127.

8. A circular disc N of radius R (Fig. 127) produces on a thin steel strip of thickness h an attraction of q lbs. per sq. in. uniformly distributed. Determine the length l of the unsupported part AC of the strip and the maximum stress in it if $h = 0.01$ inch, $R = 3$ in., and $q = 15$ lbs. per sq. in.

Solution. The length of the unsupported part of the strip can be determined from the condition that at the point C the curvature produced by the uniformly distributed load q must be equal to

$1/R$. Therefore

$$\frac{ql^2}{2} = \frac{EI_z}{R},$$

from which

$$l = \sqrt{\frac{2EI_z}{qR}} = 1/3 \text{ in.}$$

The maximum stress is determined from the equation $\sigma_{\max} = Eh/2R = 50,000$ lbs. per sq. in.

9. Determine the deflections of the cantilever beams shown in Fig. 68, assuming that the material is steel, the depth of each beam is 10 in., and the maximum bending stress is 16,000 lbs. per sq. in.

36. Deflection of a Simply Supported Beam by the Area-Moment Method.—Let us consider the case of a simply supported beam with a load P applied at point F , Fig. 128. The bending moment diagram is the triangle $a_1b_1f_1$.

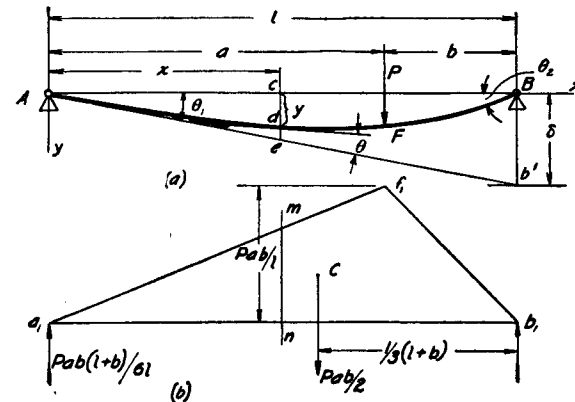


FIG. 128.

Its area is $Pab/2$, and its centroid C is at distance $(l+b)/3$ from the vertical Bb_1 . The distance δ from the end B to the line Ab' which is tangent to the deflection curve at A is obtained from equation (93) and is

$$\delta = \frac{1}{EI_z} \frac{Pab}{2} \times \frac{l+b}{3} = \frac{Pab(l+b)}{6EI_z}.$$

By using this value the slope θ_1 at the left end of the beam is found to be

$$\theta_1 = \frac{\delta}{l} = \frac{Pab(l+b)}{6lEI_z}, \quad (a)$$

which coincides with previously obtained formula (88).⁷ A simple interpretation of the formula (a) is obtained if we consider a_1b_1 as a simply supported beam, carrying the triangular load represented by the triangle $a_1f_1b_1$. The reaction at the left support a_1 of this imaginary beam is evidently

$$R = \frac{Pab}{2} \times \frac{l+b}{3} \times \frac{1}{l} = \frac{Pab(l+b)}{6l}.$$

By comparing this with formula (a), it can be concluded that the slope θ_1 is equal to the reaction at the corresponding support of the imaginary beam divided by the flexural rigidity of the real beam. The slope θ_2 at the right end of the beam can be obtained in a similar way; to get the correct sign for θ_2 the reaction at the right end must be taken with minus sign, which represents the shearing force at the right end of the imaginary beam. The imaginary beam a_1b_1 which carries the fictitious load represented by the area of the bending moment diagram is called a *conjugate beam*. It can thus be concluded that the numerical values of the slopes at the ends of a simply supported beam can be obtained by dividing the reactions at the ends of the conjugate beam by the flexural rigidity EI_z . This conclusion, which was derived for the case of a single load, holds for any transverse loading, since, as has been shown (p. 146) the moments and the deflections in the case of several loads can be obtained by the superposition of the moments and deflections due to single loads.

To calculate the slope at any point d of the deflection curve, Fig. 128, it is necessary to subtract the angle θ between the tangents at A and at d from the angle θ_1 at the support. Using equation (92) for the calculation of the angle θ , we obtain

⁷ Note that $a = l - b$.

$$\frac{dy}{dx} = \theta_1 - \theta = \frac{1}{EI_z} (R - \Delta a_1mn).$$

The first term in the parenthesis is the reaction at the left support of the conjugate beam a_1b_1 and the second is the load on the conjugate beam to the left of the cross section mn . The expression in the parenthesis therefore represents the shearing force at the cross section mn of the conjugate beam. Consequently the slope of the actual beam at a point d can be obtained by dividing the shearing force at the corresponding cross section of the conjugate beam by the flexural rigidity EI_z .

Considering next the deflection y at a point d , it may be seen from Fig. 128 that

$$y = \overline{ce} - \overline{de}. \quad (b)$$

From the triangle Ace we obtain the relation

$$\overline{ce} = \theta_1 x = \frac{Rx}{EI_z} \quad (c)$$

where R is the reaction at the left support of the conjugate beam. The second term on the right side of equation (b) represents the distance of the point d of the deflection curve from the tangent Ae and is obtained from equation (93) as

$$\overline{de} = \frac{1}{EI_z} \text{area } \Delta a_1mn \times \frac{x}{3}. \quad (d)$$

Substituting expressions (c) and (d) in equation (b), we obtain

$$y = \frac{1}{EI_z} \left(Rx - \Delta a_1mn \times \frac{x}{3} \right). \quad (e)$$

The expression in parenthesis is seen to be the bending moment at the cross section mn of the conjugate beam. Thus the deflection at any point of a simply supported beam is obtained by dividing the bending moment at the corresponding cross section of the conjugate beam by the flexural rigidity EI_z . Substituting the actual value of R in equation (e) and noting that

$$\text{area } \triangle a_1mn = \frac{Pbx^2}{2l},$$

we obtain

$$y = \frac{1}{EI_z} \left[\frac{Pabx(l+b)}{6l} - \frac{Pbx^3}{6l} \right] = \frac{Pbx}{6lEI_z} (l^2 - b^2 - x^2).$$

This checks with equation (86), which was previously obtained by integration of the differential equation of the deflection curve. The deflection for a point to the right of the load P can be calculated in a similar manner. The result will, of course, be the same as equation (87).

Having the deflection curve produced by a single load P , the deflection curve produced by any system of transverse concentrated loads can readily be obtained by employing the method of superposition. It is simply necessary to use equations (86) and (87) for each individual load.

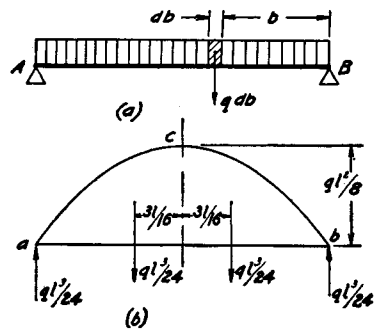


FIG. 129.

ment of load qdb shown in the Fig. 129 is

$$d\theta_1 = \frac{qabdb(l+b)}{6lEI_z} = \frac{qb(l^2 - b^2)db}{6lEI_z}.$$

The slope θ_1 produced by the total load is then the summation of the increments of slope produced by all the elements qdb from $b = 0$ to $b = l$. Thus

$$\theta_1 = \int_0^l \frac{qb(l^2 - b^2)db}{6lEI_z} = \frac{ql^3}{24EI_z}. \quad (f)$$

The deflection at the middle is obtained from equation (91), which was derived on the assumption that the load is to the right of the middle. Any element of load qdb to the right of the middle produces at the middle a deflection

$$(dy)_{x=l/2} = \frac{qbdb}{48EI_z} (3l^2 - 4b^2).$$

Summing up the deflections produced by all such elements of load to the right of the middle, and noting that the load on the left half of the beam produces the same deflection at the middle as the load on the right half, we obtain for the total deflection

$$\delta = (y)_{x=l/2} = 2 \int_0^{l/2} \frac{qbdb}{48EI_z} (3l^2 - 4b^2) = \frac{5}{384} \frac{ql^4}{EI_z}. \quad (g)$$

The results (f) and (g) coincide with formulas (83) and (82) previously obtained by integration of the differential equation of the deflection curve.

The same results are readily obtained by considering the conjugate beam ab , Fig. 129b, loaded by the parabolic segment acb , which is the bending moment diagram in this case. The total fictitious load on the conjugate beam is

$$\frac{2}{3} \times \frac{ql^2}{8} \times l,$$

and each reaction is equal to $ql^3/24$. The slope (f) is then obtained by dividing this reaction by EI_z . To calculate the deflection at the middle we find the bending moment at the middle $\bar{d}$ of the conjugate beam, which is

$$\frac{ql^3}{24} \left(\frac{l}{2} - \frac{3l}{16} \right) = \frac{5ql^4}{384}.$$

The deflection (g) is then obtained by dividing this moment by EI_z .

The method of superposition is especially useful if the distributed load covers only a part of the span as in Fig. 130. Using the expression developed above for



FIG. 130.

$(dy)_{x=l/2}$, the deflection produced at the middle by the load to the right of the middle is

$$\delta_1 = \int_a^{l/2} \frac{q b d b}{48 E I_z} (3l^2 - 4b^2).$$

The load to the left of the middle produces the deflection

$$\delta_2 = \int_c^{l/2} \frac{q b d b}{48 E I_z} (3l^2 - 4b^2).$$

The total deflection at the middle is therefore

$$\delta = \delta_1 + \delta_2 = \int_a^{l/2} \frac{q b d b}{48 E I_z} (3l^2 - 4b^2) + \int_c^{l/2} \frac{q b d b}{48 E I_z} (3l^2 - 4b^2).$$

In the case of a simply supported beam AB with a couple M acting at the end, Fig.

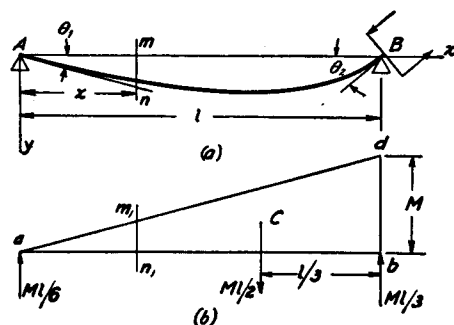


FIG. 131.

131, the bending moment diagram is a triangle abd , as shown in Fig. 131b. Considering ab as the conjugate beam, the total fictitious load is $Ml/2$. The reactions at the ends of the conjugate beam are thus $Ml/6$ and $Ml/3$. Hence the numerical values of the slopes

at the ends of the actual beam are

$$\theta_1 = \frac{Ml}{6EI_z} \quad (103)$$

and

$$\theta_2 = \frac{Ml}{3EI_z}. \quad (104)$$

The sign of the slope at the right end is of course negative. The deflection at a cross section mn of the beam is obtained by dividing the bending moment at the corresponding cross section m_1n_1 of the conjugate beam by EI_z , which gives

$$y = \frac{1}{EI_z} \left(\frac{Ml}{6} x - \frac{Ml}{2} \cdot \frac{x^2}{l^2} \cdot \frac{x}{3} \right) = \frac{Mlx}{6EI_z} \left(1 - \frac{x^2}{l^2} \right). \quad (105)$$

Problems

1. Determine the angles at the ends and the deflection under the loads and at the middle of the beam shown in Fig. 132.

Solution. The conjugate beam will be loaded by the trapezoid $adeb$, the area of which is $Pc(l-c)$. The angles at the ends are

$$\theta_1 = \theta_2 = \frac{1}{EI_z} \frac{Pc(l-c)}{2}$$

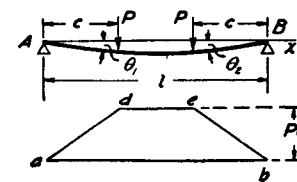


FIG. 132.

The deflection under the loads is

$$(y)_{x=c} = \frac{1}{EI_z} \left[\frac{Pc^2(l-c)}{2} - \frac{Pc^2}{2} \cdot \frac{c}{3} \right] = \frac{Pc^2}{EI_z} \left(\frac{l}{2} - \frac{2}{3}c \right).$$

The deflection at the middle, from eq. (91), is

$$(y)_{x=l/2} = \frac{Pc}{24EI_z} (3l^2 - 4c^2).$$

2. Determine the slope at the ends of the beam shown in Fig. 88.

Answer.

$$\left(\frac{dy}{dx} \right)_{x=0} = \frac{7}{180} \frac{Wl^2}{EI_z}; \quad \left(\frac{dy}{dx} \right)_{x=l} = -\frac{8}{180} \frac{Wl^2}{EI_z}.$$

3. Determine the deflection at the middle of the beam AB , shown in Fig. 133, when $I_z = 91.4 \text{ in.}^4$, $q = 500 \text{ lbs. per foot}$, $l = 24 \text{ feet}$, $a = 12 \text{ feet}$, $b = 8 \text{ feet}$. $E = 30 \times 10^6 \text{ lbs. per sq. in.}$

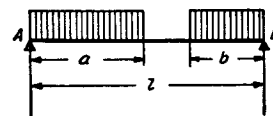


FIG. 133.

Solution. Due to the fact that $a = l/2$ the deflection produced at the middle by the load acting on the left half of the beam, from eq. (82), is

$$(y_1)_{x=l/2} = \frac{1}{2} \frac{5}{384} \frac{q l^4}{EI_z}.$$

The deflection produced at the middle by the load on the right

half of the beam is

$$(y_2)_{x=l/2} = \int_0^b \frac{qcdc}{48EI_z} (3l^2 - 4c^2) = \frac{25}{48 \times 162} \times \frac{ql^4}{EI_z}.$$

The total deflection is

$$(y)_{x=l/2} = (y_1)_{x=l/2} + (y_2)_{x=l/2} = \left(\frac{1}{2} \frac{5}{384} + \frac{25}{48 \times 162} \right) \frac{ql^4}{EI_z} = 1.02 \text{ in.}$$

4. Determine the deflection at the middle of the beam shown in Fig. 91 when the load is in a position to produce the maximum bending moment.

Suggestion. The deflection can be obtained by using eq. (91) together with the method of superposition and substituting $b = l/2 - d/4$ in this equation for one load and $b = l/2 - \frac{3}{4}d$ for the other.

5. Determine the deflections at the middle and the angles of

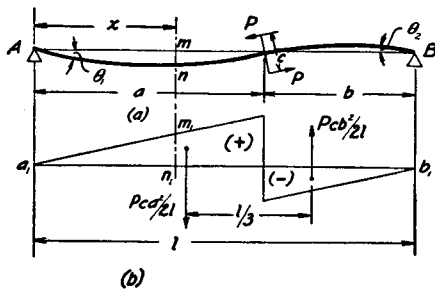


FIG. 134.

Solution. The loading of the conjugate beam is indicated in Fig. 134 (b). The reactions at a_1 and b_1 are

$$R_a = \frac{1}{l} \left[\frac{Pca^2}{2l} \left(b + \frac{a}{3} \right) - \frac{Pcb^2}{2l} \cdot \frac{2}{3} b \right];$$

$$R_b = \frac{1}{l} \left[\frac{Pca^2}{2l} \frac{2a}{3} - \frac{Pcb^2}{2l} \left(a + \frac{b}{3} \right) \right].$$

Therefore

$$\theta_1 = \frac{Pc}{2l^2EI_z} \left[a^2 \left(b + \frac{a}{3} \right) - \frac{2}{3} b^3 \right] = \frac{Pc}{2lEI_z} \left(\frac{l^2}{3} - b^2 \right);$$

$$\theta_2 = -\frac{Pc}{2l^2EI_z} \left[\frac{2}{3} a^3 - b^2 \left(a + \frac{b}{3} \right) \right] = \frac{Pc}{2lEI_z} \left(\frac{l^2}{3} - a^2 \right).$$

rotation of the ends of the beams shown in Figs. 67 (b) and 67 (e). Assume in these calculations a standard I beam of 8 in. depth and 5.97 sq. in. area, $E = 30 \times 10^6$ lbs. per sq. in.

6. Determine the angles θ_1 and θ_2 and the deflection at any cross section mn of a beam simply supported at the ends and bent by a couple Pc (Fig. 134).

If $a = b = l/2$, we obtain

$$\theta_1 = \theta_2 = \frac{Pcl}{24EI_z}.$$

If $a > l/\sqrt{3}$ the angle θ_2 changes its sign. The bending moment at the cross section m_1n_1 of the conjugate beam is

$$A_1x - \frac{Pca^2}{2l} \frac{x^2}{2} = \frac{Pcx}{2l^2} \left[a^2 \left(b + \frac{a}{3} \right) - \frac{2}{3} b^3 \right] - \frac{Pcx^3}{6l}.$$

Therefore the deflection curve for the left part of the actual beam is

$$y = \frac{Pcx}{2l^2EI_z} \left[a^2 \left(b + \frac{a}{3} \right) - \frac{2}{3} b^3 \right] - \frac{Pcx^3}{6lEI_z}.$$

7. A beam with supported ends is bent by two couples M_1 and M_2 , applied at the ends (Fig. 135). Determine the angles of rotation of the ends and the position of the cross section in which the deflection is a maximum.

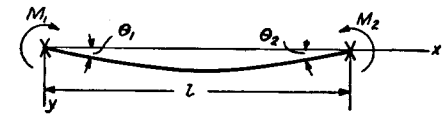


FIG. 135.

Solution. The absolute values of the angles are

$$\theta_1 = \frac{M_1l}{3EI_z} + \frac{M_2l}{6EI_z}; \quad \theta_2 = \frac{M_2l}{3EI_z} + \frac{M_1l}{6EI_z}.$$

The deflection curve, by using eq. (105), is

$$y = \frac{M_1l(l-x)}{6EI_z} \left[1 - \left(\frac{l-x}{l} \right)^2 \right] + \frac{M_2lx}{6EI_z} \left(1 - \frac{x^2}{l^2} \right).$$

The position of maximum deflection can be found from this equation by equating the first derivative to zero.

8. A beam is bent by two couples as shown in Fig. 136. Determine the ratio $M_1 : M_2$ if the point of inflection is at a distance $l/3$ from the left support.

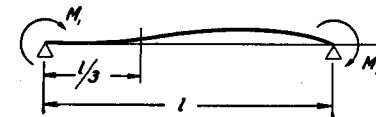


FIG. 136.

Answer. $M_2 = 2M_1$.

9. Two planks of different thicknesses h_1 and h_2 , resting one on the other, support a uniformly distributed load as shown in Fig. 137. Determine the ratio of the maximum stresses occurring in each.

Solution. Both planks have the same deflection curves and curvature; hence their bending moments are in the same ratio as the moments of inertia of their cross sections, i.e., in the ratio $h_1^3 : h_2^3$. The section moduli are in the ratio $h_1^2 : h_2^2$; hence the maximum stresses are in the ratio $h_1 : h_2$.

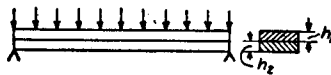


FIG. 137.

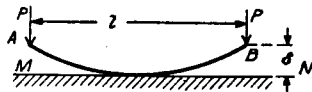


FIG. 138.

10. A steel bar AB has such an initial curvature that after being straightened by the forces P (Fig. 138) it produces a uniformly distributed pressure along the length of the rigid plane surface MN . Determine the forces P necessary to straighten the bar and the maximum stress produced in it if $l = 20$ in., $\delta = 0.1$ in., and the cross section of the bar is a square having 1 in. sides.

Solution. To obtain a uniformly distributed pressure, the initial curvature of the bar must be the same as the deflection curve of a simply supported beam carrying a uniformly distributed load of intensity $2P/l$. Then we obtain

$$M_{\max} = \frac{2Pl^2}{l^3} = \frac{Pl}{4}, \quad (a)$$

$$\delta = \frac{5}{384} \times \frac{2P}{l} \times \frac{l^4}{EI_z}. \quad (b)$$

The maximum stress will be

$$\sigma_{\max} = \frac{M_{\max}}{Z} = \frac{Plh}{8I_z}. \quad (c)$$

Now from (b) and (c)

$$\sigma_{\max} = \frac{24E\delta h}{5l^2} = \frac{24 \times 30 \times 10^6 \times 0.1 \times 1}{5 \times 20^2} = 36,000 \text{ lbs. per sq. in.}$$

and from (c)

$$P = 1,200 \text{ lbs.}$$

37. Deflection of Beams with Overhangs.—A beam with an overhang can be divided into two parts: the one between the supports which is to be treated as a beam with supported ends and the overhang which is to be treated as a cantilever. As an

illustration, we consider the bending of a beam with an overhang under the action of a uniformly distributed load q (Fig. 139). The beam is divided into the parts AB and BC and the action of the overhang on the portion of the beam between the supports is replaced by a shearing force qa and a couple $M = qa^2/2$. We find that the shearing force is directly transmitted to the sup-

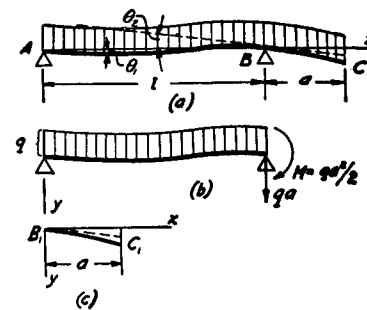


FIG. 139.

port and that only the couple $qa^2/2$ need be considered. Then the deflection at any cross section between the supports is obtained by subtracting the deflection produced by the couple $qa^2/2$ from the deflections produced by the uniform load q (Fig. 139, b). Using eqs. (81) and (105), we obtain

$$y = \frac{q}{24EI_z} (l^3x - 2lx^3 + x^4) - \frac{qa^2lx}{12EI_z} \left(1 - \frac{x^2}{l^2} \right).$$

The angle of rotation of the cross section at B is obtained by using eqs. (83) and (104), from which, by considering rotation positive when in the clockwise direction, we have

$$\theta_2 = \frac{qa^2l}{6EI_z} - \frac{ql^3}{24EI_z}.$$

The deflection at any cross section of the overhang (Fig. 139, c) is now obtained by superposing the deflection of a cantilever (eq. 101) on the deflection

$$\theta_2x = \left(\frac{qa^2l}{6EI_z} - \frac{ql^3}{24EI_z} \right) x$$

due to the rotation of the cross section B .

Problems

1. Determine the deflection and the slope at the end C of the beam shown in Fig. 141a.

Answer. Deflection = $\frac{Pa^2(l+a)}{3EI_z}$; Slope = $\frac{Pa(2l+3a)}{6EI_z}$.

2. For the beam shown in Fig. 140 determine the deflection at the end C and also that at the midpoint between the supports.

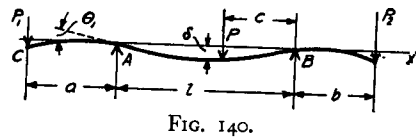


FIG. 140.

Solution. That part of the beam between the supports will be in the condition of a beam loaded by the force P and by the couples P_1a and P_2b at the supports. By using

eqs. (91) and (105), pp. 143, 159, and the method of superposition, the deflection at the middle is

$$\delta = \frac{Pc}{48EI_z} (3l^2 - 4c^2) - \frac{P_1al^2}{16EI_z} - \frac{P_2bl^2}{16EI_z}.$$

The angle θ_1 at the support A is obtained from eqs. (88), (103) and (104) on pp. 142, 158,

$$\theta_1 = \frac{Pc(l^2 - c^2)}{6EI_z} - \frac{P_1al}{3EI_z} - \frac{P_2bl}{6EI_z}.$$

From eq. (95) the deflection at the end C is

$$\frac{P_1a^3}{3EI_z} - a\theta_1.$$

3. A beam with an overhang is bent in one case by the force P at the end (Fig. 141, a), and in another case by the same force applied at the middle of the span (Fig. 141, b). Prove that the deflection at the point D in the first case is equal to the deflection at the end C in the second case.

Answer. In each case the deflection is

$$\frac{P^2a}{16EI_z}.$$

4. A beam of length l with two equal overhangs is loaded by two equal forces P at the ends (Fig. 142). Determine the ratio x/l

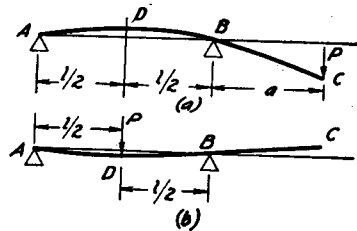


FIG. 141.

at which (1) the deflection at the middle is equal to the deflection at either end, (2) the deflection at the middle has its maximum value.

Answer. (1) $x = 0.152l$; (2) $x = l/6$.

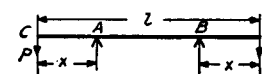


FIG. 142.

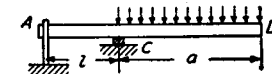


FIG. 143.

5. A wooden beam of circular cross section supported at C , with the end attached at A , carries a uniformly distributed load q on the overhang CD (Fig. 143). Determine the diameter of the cross section and the deflection at D if $l = 3$ feet, $a = 6$ feet, $q = 300$ lbs. per foot, $\sigma_w = 1,200$ lbs. per sq. in.

Solution. The diameter d is found from equation

$$\frac{qa^2}{2} : \frac{\pi d^3}{32} = \sigma_w.$$

Then the deflection at the end D is

$$\delta = \frac{qa^4}{8EI_z} + \frac{qa^3l}{6EI_z}.$$

6. A beam of length l carries a uniformly distributed load of intensity q (Fig. 144). Determine the length of overhangs to make the numerical maximum value of the bending moment as small as possible. Determine the deflection at the middle for this condition.

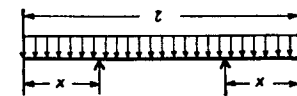


FIG. 144.

Solution. Making the numerical values of the bending moments at the middle and at the supports equal we obtain

$$x = 0.207l.$$

The deflection at the middle is determined from the equation

$$\delta = \frac{5}{384} \cdot \frac{q(l-2x)^4}{EI_z} - \frac{qx^2(l-2x)^2}{16EI_z},$$

in which the first term on the right side represents the deflection produced by the load between the support (eq. 82) and the second, the deflection produced by the load on the overhangs (eq. 105).

7. Determine the deflections at the ends of the overhangs for the beams represented in Fig. 74 a , b , c . Assume a standard I beam of 8 in. depth and 5.97 sq. in. area. $E = 30 \times 10^6$ lbs. per sq. in.

38. The Deflection of Beams When the Loads Are Not Parallel to One of the Two Principal Planes of Bending.—

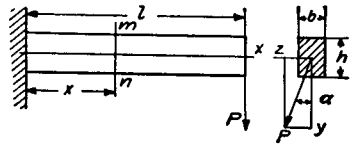


FIG. 145.

Let us consider first a simple example of a cantilever, whose cross section has two axes of symmetry (Fig. 145). The load P at the end is perpendicular to the axis of the beam and makes an angle α with the principal axis y of the cross section. In calculating the stresses and deflections of the beam the method of superposition will be used. The load P will be resolved into two components $P \cos \alpha$ and $P \sin \alpha$ in the directions of the two principal axes of the cross section. The deflection produced by each of these components can easily be obtained by using the theory of bending in the plane of symmetry. Then the resultant deflection is obtained by superposition. The absolute values of the two components of the bending moment at any cross section mn of the cantilever are $M_z = P \cos \alpha(l - x)$ about the z axis, and $M_y = P \sin \alpha(l - x)$ about the y axis. From the directions of the two components and of the axes y and z , it will be seen that the moment M_z produces compression in points with a positive y and M_y produces compression in points with a positive z . Then the normal stress σ_x at any point of the cross section (y, z) is obtained by adding together the stresses produced by M_z and M_y separately. We thus arrive at the equation:

$$\sigma_x = -\frac{P \cos \alpha(l - x)y}{I_z} - \frac{P \sin \alpha(l - x)z}{I_y}$$

$$= -P(l - x) \left[\frac{y \cos \alpha}{I_z} + \frac{z \sin \alpha}{I_y} \right]. \quad (a)$$

The neutral axis is found by taking points with such coordinates that the expression in brackets in eq. (a) equals zero. The equation of the neutral axis is therefore

$$\frac{y \cos \alpha}{I_z} + \frac{z \sin \alpha}{I_y} = 0. \quad (b)$$

This is an axis through the centroid of the cross section making an angle β with the z axis (see Fig. 146) such that, from eq. (b),

$$\tan \beta = -\frac{y}{z} = \tan \alpha \frac{I_z}{I_y}. \quad (c)$$

It will be seen that, in general, $\tan \beta$ is not equal to $\tan \alpha$; hence the neutral axis nn is not perpendicular to the plane of the bending forces and the plane of the deflection curve, which is perpendicular to nn , does not coincide with the plane of the bending forces. These two planes coincide only when $\tan \alpha$

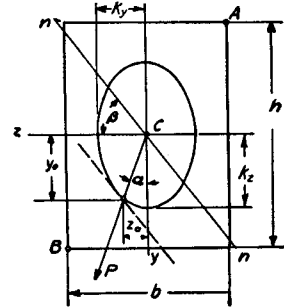


FIG. 146.

$= 0$ or ∞ or $I_z = I_y$. In the first two cases the plane of the bending forces coincides with one of the principal planes of bending. In the last case the ellipse of inertia becomes a circle since the two principal moments of inertia are equal and any two perpendicular directions may be taken as the two principal axes of the cross section. When I_z/I_y is a large number, i.e., when the rigidity of the beam in the xy plane is much larger than that in the xz plane, $\tan \beta$ becomes large in comparison with $\tan \alpha$ and when angle α is small, angle β will approach 90° and the neutral axis will approach the vertical axis. The deflection will be principally in the xz plane, i.e., there is a tendency to deflect in the plane of greatest flexibility. This can be demonstrated in a very simple manner on a thin rule. The slightest deviation of the bending force from the plane of greatest rigidity results in a bending in the perpendicular direction. This can be shown also by resolving the force P (Fig. 145) into two components and calculating deflections produced by each component. If the flexural rigidity of the cantilever in the horizontal plane is very small in comparison with the rigidity in vertical plane, a small horizontal component may produce a much greater horizontal deflection than the deflection in the vertical plane; hence the resultant deflection will be principally in the plane of

greatest flexibility. It is interesting to note that the neutral axis nn is parallel to the tangent drawn to the ellipse of inertia at the point of intersection of this ellipse with the direction of the force P . This can be proved as follows. The equation of the ellipse is

$$\frac{y^2}{k_z^2} + \frac{z^2}{k_y^2} = 1$$

and the equation of the tangent at the point with the coordinates y_0 and z_0 (Fig. 146) is

$$\frac{yy_0}{k_z^2} + \frac{zz_0}{k_y^2} = 1.$$

The tangent of the angle between the z axis and this tangent will be

$$\frac{z_0}{y_0} \cdot \frac{k_z^2}{k_y^2} = \tan \alpha \frac{I_z}{I_y} = \tan \beta.$$

When the direction of the neutral axis is determined the points of maximum normal stress will be those most distant from it. In our case the maximum tension will be at point A and maximum compression at point B . Substituting in eq. (a) $x = 0$; $y = -(h/2)$; $z = -(b/2)$, we obtain

$$(\sigma_x)_{\max} = Pl \left(\frac{h \cos \alpha}{2I_z} + \frac{b \sin \alpha}{2I_y} \right). \quad (d)$$

The compressive stresses at the point B will have the same magnitude. The method developed above for the case of a cantilever with two planes of symmetry and loaded at the end can be applied also to beams supported at the ends and loaded by several loads. Resolving each force into two components parallel to the two axes of symmetry of the cross section, the problem is reduced to two simple problems of bending of a beam in the two principal planes. The resultant deflections will be obtained by superposing the two deflections in the principal planes.

Problems

1. A cantilever beam of Z section is loaded at the end by a vertical load $P = 400$ lbs. (Fig. 147). Determine the maximum normal stress σ_x and the vertical and horizontal components of the deflection at the end. The dimensions are as indicated in the figure. $\alpha = 17^\circ 20'$, principal moments of inertia

$$I_{z_1} = 60.3 \text{ in.}^4; \quad I_{y_1} = 3.54 \text{ in.}^4$$

Answer. $(\sigma_x)_{\max} = 6,420$ lbs. per sq. in. at B ; $\delta_{\text{vert.}} = 0.178$ in.; $\delta_{\text{horiz.}} = 0.336$ in.

2. A cantilever of rectangular cross section is bent by a force P at the end. What curve will be described by the loaded end, when the angle α (Fig. 145) varies from 0 to 2π ?

Answer. The curve will be an ellipse with the semi axes

$$\frac{Pl^3}{3EI_z} \quad \text{and} \quad \frac{Pl^3}{3EI_y}.$$

3. A wooden beam of rectangular cross section carries a uniformly distributed load of intensity q and is supported at the ends in the position shown in Fig. 148. Determine the maximum normal stress and the vertical deflection at the middle if the length of the beam $l = 10$ feet, $q = 200$ lbs. per foot, $h = 8$ in., $b = 6$ in., $\tan \alpha = 1/3$.

Solution. The maximum bending moment will be at the middle

$$M_{\max} = \frac{ql^2}{8} = \frac{200 \times 10^2}{8} = 2,500 \text{ lbs. feet} = 30,000 \text{ lbs. ins.}$$

The components of the bending moment in the principal planes are $M_z = M_{\max} \cos \alpha = 30,000 \times 0.949 = 28,500$ lbs. ins. and $M_y = M_{\max} \sin \alpha = 30,000 \times 0.316 = 9,480$ lbs. in. The maximum stress at the point B is

$$(\sigma_x)_{\max} = \frac{6 \times 28,500}{bh^2} + \frac{6 \times 9,480}{hb^2} = 643 \text{ lbs. per sq. in.}$$

The deflections at the middle in the two principal planes are

$$\delta_y = \frac{5}{384} \frac{q l^4 \cos \alpha}{EI_z} \quad \text{and} \quad \delta_z = \frac{5}{384} \frac{q l^4 \sin \alpha}{EI_y}.$$

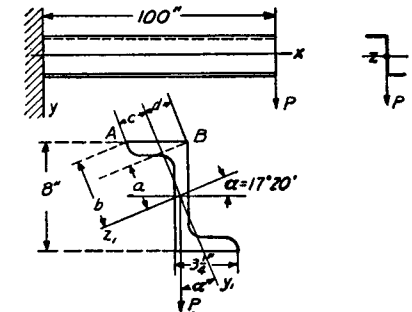


FIG. 147.

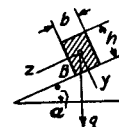


FIG. 148.

The vertical deflection at the middle is

$$\delta = \delta_y \cos \alpha + \delta_z \sin \alpha = \frac{5}{384} \frac{q l^4}{EI_z} \left(\cos^2 \alpha + \frac{I_z}{I_y} \sin^2 \alpha \right) \\ = 0.117 \text{ in.} \times 1.08 = .126 \text{ in.}$$

4. Solve the above problem if the distance between the supports is 6 feet and the beam has two equal overhangs each 2 feet long.

39. Effect of Shearing Force on the Deflection of Beams.—In the previous discussion (see p. 134) only the action of the bending moment in causing deflection was considered. An additional deflection will be produced by the shearing force, in the form of a mutual sliding of adjacent cross sections along each other. As a result of the non-uniform distribution of the shearing stresses, the cross sections, previously plane, become curved as in Fig. 149, which shows the bending due to shear alone.⁸ The elements of the cross sections at the centroids remain vertical and slide along one another; therefore the slope of the deflection

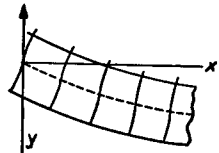


FIG. 149.

curve, due to the shear alone, is equal at each cross section to the shearing strain at the centroid of this cross section. Denoting by y_1 the deflections due to shear, we obtain for any cross section the following expression for the slope:

$$\frac{dy_1}{dx} = \frac{(\tau_{yx})_{y=0}}{G} = \frac{\alpha V}{AG}, \quad (a)$$

in which V/A is the average shearing stress τ_{yx} , G is the modulus in shear and α is a numerical factor with which the average shearing stress must be multiplied in order to obtain the shearing stress at the centroid of the cross sections. For a rectangular cross-section $\alpha = 3/2$ (see eq. 66, p. 114); for a circular cross-section $\alpha = 4/3$ (see eq. 68, p. 118). With a continuous load on the beam, the shearing force V is a continuous function which may be differentiated with respect to x . The curvature caused by the shear alone is then

$$\frac{d^2 y_1}{dx^2} = \frac{\alpha}{AG} \frac{dV}{dx} = -\frac{\alpha}{AG} q.$$

⁸ The deformation produced by the bending moment and consisting of a mutual rotation of adjacent cross sections is removed.

The sum of this and the curvature produced by the bending moment (see eq. 79) gives

$$\frac{d^2 y}{dx^2} = -\frac{1}{EI_z} \left(M + \frac{\alpha EI_z}{AG} q \right). \quad (106)$$

This equation must be used instead of eq. (79) to determine deflections in all cases in which the effect of the shearing force should be taken into consideration.⁹ Knowing M and q as functions of x , eq. (106) can be easily integrated in the same manner as has been shown in article 32.

The conjugate beam method (see p. 154) may also be applied to good advantage in this case by taking as ordinates of the imaginary load diagram

$$M + \alpha \frac{EI_z}{AG} q, \quad (b)$$

instead of only M .

Let us consider, for example, the case of a simply supported beam with a uniform load (Fig. 150). The bending moment at any section x is

$$M = \frac{ql}{2} x - \frac{qx^2}{2}. \quad (c)$$

The load on the conjugate beam consists of two parts: (1) that represented by the first term of (b) and given by the parabolic bending moment diagram (Fig. 150, b) and (2) that represented by the second term of (b), which is $\alpha(EI_z/AG)q$. Since q is constant, the second term is a uniformly distributed load shown in Fig. 150 (c).

The additional deflection at any section due to the shearing force is the bending moment produced at this section of the conjugate beam by such a load, divided by EI_z . At the middle of the beam the additional deflection is consequently

$$\delta_1 = \frac{1}{EI_z} \left(\alpha \frac{EI_z}{AG} q \right) \frac{l^2}{8} = \frac{\alpha l^2 q}{8AG}.$$

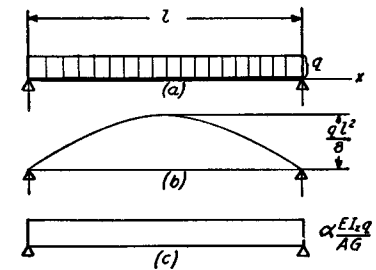


FIG. 150.

Adding this to the deflection due to the bending moment (see eq.

⁹ Another way of determining additional deflection due to shear is discussed on page 298.

82, p. 138), we obtain the total deflection

$$\delta = \frac{5}{384} \frac{ql^4}{EI_z} + \frac{\alpha l^2 q}{8AG} = \frac{5}{384} \frac{ql^4}{EI_z} \left(1 + \frac{48\alpha}{5} \frac{k_z^2 E}{l^2 G} \right), \quad (d)$$

in which $k_z = \sqrt{I_z/A}$ is the radius of gyration of the cross section with respect to the z axis.

For a rectangular cross-section of depth h , $k_z^2 = \frac{1}{12}h^2$, $\alpha = 3/2$. Putting $E/G = 2(1 + \mu) = 2.6$, we obtain from (d)

$$\delta = \frac{5}{384} \frac{ql^4}{EI_z} \left(1 + 3.12 \frac{h^2}{l^2} \right).$$

It may be seen that for $l/h = 10$ the effect of the shearing force on the deflection is about 3 per cent. As the ratio l/h decreases this effect increases.

The factor α is usually larger than 2 for I beams and when they are short the effect of the shearing force may be comparatively great. Using eq. (70) and Fig. 106, we have

$$\frac{\alpha V}{A} = \frac{V}{b_1 I_z} \left[\frac{bh^2}{8} - \frac{h_1^2}{8} (b - b_1) \right],$$

from which

$$\alpha = \frac{A}{b_1 I_z} \left[\frac{bh^2}{8} - \frac{h_1^2}{8} (b - b_1) \right]. \quad (e)$$

For example, suppose $h = 24$ in., $A = 31.0$ sq. in., $I_z = 2,810$ in.⁴, the thickness of the web $b_1 = 5/8$ in., $l = 6h$. Then eq. (e) gives $\alpha = 2.42$. Substituting in eq. (d), we find

$$\delta = \frac{5}{384} \frac{ql^4}{EI_z} \left(1 + \frac{48}{5} \times 2.42 \times \frac{2,810}{31 \times 144^2} \times 2.6 \right) = 1.265 \frac{5ql^4}{384EI_z}.$$

The additional deflection due to shear in this case is equal to 26.5 per cent of the deflection produced by the bending moment and must therefore be considered.

In the case of a concentrated load P (Fig. 151) such a load can be considered as the limiting case of a load distributed over a very short portion ϵ of the beam. The amount of the imaginary loading P_1 on the conjugate beam A_1B_1 , corresponding to the second term in expression (b),

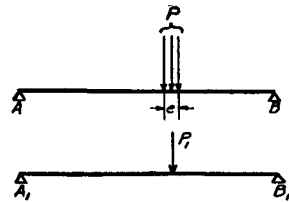


FIG. 151.

will be

$$P_1 = \alpha \frac{EI_z}{AG} P. \quad (f)$$

The additional deflection due to shearing forces is obtained by dividing by EI_z the bending moment produced in the conjugate beam by the imaginary concentrated load given by eq. (f). For instance, for central loading of a beam the bending moment at the middle of the conjugate beam produced by the load (f) will be $\alpha(EI_z/AG)Pl/4$ and the additional deflection at the middle due to shearing forces is

$$\delta_1 = \frac{\alpha}{AG} \frac{Pl}{4}. \quad (g)$$

Adding this to the deflection produced by the bending moment alone (see eq. 90, p. 143), the following expression for the complete deflection is obtained:

$$\delta = \frac{Pl^3}{48EI_z} + \frac{\alpha}{AG} \cdot \frac{Pl}{4} = \frac{Pl^3}{48EI_z} \left(1 + \frac{12\alpha k_z^2 E}{l^2 G} \right).$$

For a beam of rectangular cross-section of depth h we have

$$\frac{k_z^2}{l^2} = \frac{h^2}{12l^2}; \quad \alpha = \frac{3}{2},$$

and we obtain

$$\delta = \frac{Pl^3}{48EI_z} \left(1 + 3.90 \frac{h^2}{l^2} \right). \quad (h)$$

For $h/l = 1/10$ the additional effect of the shearing force is about 4 per cent.

It has been assumed throughout the above discussion that the cross sections of the beam can warp freely as shown in Fig. 149. The uniformly loaded beam is one case in which this condition is approximately satisfied. The shearing force at the middle of such a beam is zero and there will be no warping here. The warping increases gradually with the shearing force as we proceed along the beam to the left or to the right of the middle. The condition of symmetry of deformation with respect to the middle section is therefore satisfied. Consider now bending by a concentrated load at the middle. From the condition of symmetry the middle cross section of the beam must remain plane. At the same time, adjacent cross sections to the right and to the left of the load carry a shearing force equal to $P/2$, and warping of cross sections caused by these shearing forces should take place. From

the condition of continuity of deformation, however, there can be no abrupt change from a plane middle section to warped adjacent sections. There must be a continuous increase in warping as we proceed along the beam in either direction from the middle, and only at some distance from the load can the warping be such as a shearing force $P/2$ produces under conditions of freedom in warping. From this discussion it must be concluded that in the neighborhood of the middle cross section the stress distribution will not be that predicted by the elementary theory of bending (see p. 113). Warping will be partially prevented and the additional deflection due to shearing forces will be somewhat less than that found above (see eq. g). A more detailed investigation¹⁰ shows that in the case of a concentrated load at the middle the deflection at the middle is

$$\delta = \frac{Pl^3}{48EI} \left[1 + 2.85 \frac{h^2}{l^2} - 0.84 \left(\frac{h}{l} \right)^3 \right]. \quad (k)$$

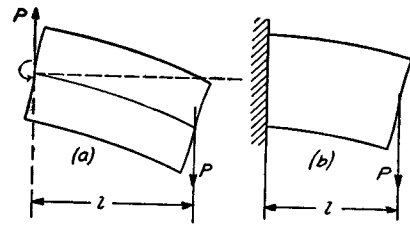


FIG. 152.

We have an analogous condition also in the case of a cantilever beam. If the built-in cross section can warp freely as shown in Fig. 152 (a), the conditions will be as assumed in the derivation of eq. (h). The deflection of a cantilever of rectangular cross section will be obtained

by substituting l for $l/2$ and P for $P/2$ in this equation, giving

$$\delta = \frac{Pl^3}{3EI} \left(1 + 0.98 \frac{h^2}{l^2} \right). \quad (l)$$

When the built-in cross section is completely prevented from warping (Fig. 152, b), the conditions will be the same as assumed in the derivation of eq. (k) and the deflection will be

$$\delta = \frac{Pl^3}{3EI} \left[1 + 0.71 \frac{h^2}{l^2} - 0.10 \left(\frac{h}{l} \right)^3 \right], \quad (m)$$

which is less than the deflection given by (l).

¹⁰ See L. N. G. Filon, Phil. Trans. Roy. Soc. (A), Vol. 201, p. 63, 1903, and S. Timoshenko, Phil. Mag., Vol. 47, p. 1095, 1924. See also Th. v. Kármán, Scripta Universitatis atque Bibliothecae Hierosolimitanarum, 1923, and writer's "Theory of Elasticity," p. 95, 1934.

CHAPTER VI

STATICALLY INDETERMINATE PROBLEMS IN BENDING

40. Redundant Constraints.—In our previous discussion three types of beams have been considered: (a) a cantilever beam, (b) a beam supported at the ends and (c) a beam with overhangs. In all three cases the reactions at the supports can be determined from the fundamental equations of statics; hence the problems are *statically determinate*. We will now consider problems on the bending of beams in which the equations of statics are not sufficient to determine all the reactive forces at the supports, so that additional equations, based on a consideration of the deflection of the beams, must be derived. Such problems are called *statically indeterminate*.

Let us consider the various types of supports which a beam may have. The support represented in Fig. 153 (a) is

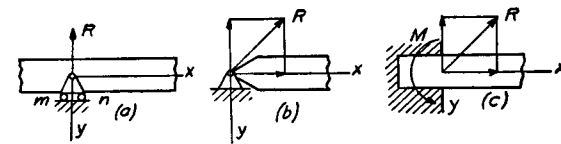


FIG. 153.

called a *hinged movable support*. Neglecting the friction in the hinge and in the rollers, it is evident that in this type of support the reaction must act through the center of the hinge and must be perpendicular to the plane mn on which the rollers are moving. Hence we know the point of application of the reaction and its direction. There remains only one unknown element, the magnitude of the reaction.

In Fig. 153 (b) a *hinged immovable* support is shown. In this case the reaction must go through the center of the hinge but it may have any direction in the plane of the figure. We have two unknowns to determine from the equations of

statics, the direction of the reaction and its magnitude, or, if we like, the vertical and horizontal components of the reaction.

In Fig. 153 (c) a *built-in end* is represented. In this case not only the direction and the magnitude of the reaction are unknown, but also the point of application. The reactive forces distributed over the built-in section can be replaced by a force R applied at the centroid of the cross section and a couple M . We then have three unknowns to determine from the equations of statics, the two components of the reactive force R and the magnitude of the couple M .

For beams loaded by transverse loads in one plane we have, for determining the reactions at the supports, the three equations of statics, namely,

$$\Sigma X = 0; \quad \Sigma Y = 0; \quad \Sigma M = 0. \quad (a)$$

If the beam is supported so that there are only three unknown *reactive elements*, they can be determined from eqs. (a); hence the problem is statically determinate. These three elements are just sufficient to assure the immovability of the beam. When the number of reactive elements is larger than three, we say there are *redundant constraints* and the problem is statically indeterminate.

A cantilever is supported at the built-in end. In this case, as was explained above, the number of unknown reactive elements is three and they can be determined from the equations of statics. For beams supported at both ends and beams with overhangs it is usually assumed that one of the supports is an immovable and the other a movable hinge. In

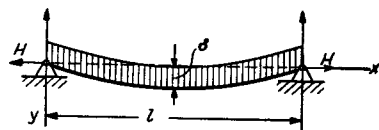


FIG. 154.

such a case we have again three unknown reactive elements, which can be determined from the equations of statics.

If the beam has immovable hinges at both ends (Fig. 154), the problem becomes statically indeterminate. At each end we have two unknown elements, the two components of the

corresponding reaction, and for determining these four unknowns we have only the three equations (a). Hence we have one redundant constraint and a consideration of the deformation of the beam becomes necessary to determine the reactions. The vertical components of the reactions can be calculated from the equations of statics. In the case of vertical loads it can be concluded also from statics that the horizontal components H are equal and opposite in direction. To find the magnitude of H let us consider the elongation of the axis of the beam during bending. A good approximation to this elongation can be obtained by assuming that the deflection curve of the beam is a parabola,¹ the equation of which is

$$y = \frac{4\delta x(l-x)}{l^2} \quad (b)$$

where δ is the deflection at the middle. The length of the curve is

$$s = 2 \int_0^{l/2} \sqrt{dx^2 + dy^2} = 2 \int_0^{l/2} dx \sqrt{1 + \left(\frac{dy}{dx}\right)^2}. \quad (c)$$

In the case of a flat curve the quantity $(dy/dx)^2$ is small in comparison with unity and neglecting small quantities of order higher than the second we obtain approximately

$$\sqrt{1 + \left(\frac{dy}{dx}\right)^2} \approx 1 + \frac{1}{2} \left(\frac{dy}{dx}\right)^2.$$

Substituting this expression in equation (c) and using equation (b) we find the length of the curve to be

$$s = l \left(1 + \frac{8}{3} \frac{\delta^2}{l^2} \right).$$

Then the difference between the length of the curve and the distance l between the supports represents the total axial elongation of the beam and is $(8/3)(\delta^2/l)$; the unit elongation

¹ The exact expression for the deflection curve will be given later (see Part II).

is then $(8/3)(\delta^2/l^2)$. Knowing this and denoting by E the modulus of elasticity of the material of the beam and by A its cross-sectional area, we obtain the horizontal reaction from equation:

$$H = \frac{8}{3} \frac{\delta^2}{l^2} EA. \quad (b)$$

It is interesting to note that for most beams in practice the deflection δ is very small in comparison with the length and the tensile stress $(8/3)(\delta^2/l^2)E$ produced by the forces H is usually small in comparison with bending stresses and can be neglected. This justifies the usual practice of calculating beams with supported ends by assuming that one of the two supports is a movable hinge, although the special provisions for permitting free motion of the hinge are actually used only in cases of large spans such as bridges.

In the case of the bending of flexible bars and thin metallic strips, where the deflection δ is no longer very small in comparison with l , the tensile stresses produced by the longitudinal forces H can not be neglected. Such problems will be discussed later (see Part II). In the following discussion of statically indeterminate problems of bending the method of superposition will be used and the solutions will be obtained by combining statically determinate cases in such a manner as to satisfy the conditions at the supports.

41. Beam Built-in at One End and Supported at the Other.—Consider first the case where there is a single concentrated load P (Fig. 155).² In this case we have three unknown reactive elements at the left end and one at the right end. Hence the problem is statically indeterminate with one redundant constraint. In the solution of this problem let us consider as redundant the constraint which prevents the left end A of the beam from rotating during bending. Removing this constraint, we obtain the statically determinate problem shown in Fig. 155 (b). The bending produced by the statically indeterminate couple M_a will be studied separately

as shown in Fig. 155 (c).² It is evident that the bending of the beam represented in Fig. 155 (a) can be obtained by a combination of cases (b) and (c).

It is only necessary to adjust the magnitude of the couple M_a at the support in such a manner as to satisfy the condition

$$\theta_1 = -\theta_1'. \quad (a)$$

Thus, the rotation of the left end of the beam, due to the force P , will be annihilated by M_a and the condition of a built-in end with zero slope will be satisfied. To obtain the statically indeterminate couple M_a it is necessary only to substitute in eq. (a) the known values for the angles θ_1 and θ_1' from equations (88), p. 142, and (104), p. 158. Then

$$\frac{Pc(l^2 - c^2)}{6lEI_z} = -\frac{M_al}{3EI_z},$$

from which

$$M_a = -\frac{Pc(l^2 - c^2)}{2l^2}. \quad (107)$$

The bending moment diagram can now be obtained by combining the diagrams for cases (b) and (c) as shown by the shaded area in Fig. 155 (d). The maximum bending moment will be either that at a or that at d .

The deflection at any point can easily be obtained by subtracting from the deflection at this point produced by the load P the deflection produced by the couple M_a . The equations of the deflection curves for both these cases have already been given in (86) and (87), p. 142, and in (105), p. 159. Let us take, for instance, the case $c < \frac{1}{2}l$ and calcu-

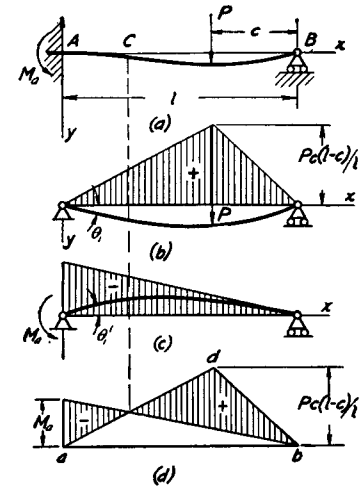


FIG. 155.

² Deflection curves and bending moment diagrams are shown together.

late the deflection at the middle of the span. From eqs. (91) and (105)

$$\delta = \frac{Pc}{48EI_z} (3l^2 - 4c^2) + \frac{M_a l^2}{16EI_z}$$

or, by using eq. (107),

$$\delta = \frac{Pc}{96EI_z} (3l^2 - 5c^2).$$

At the point *C* where the bending moment becomes zero, the curvature of the deflection curve is also zero and we have a point of inflection, i.e., a point where the curvature changes sign.

It may be seen from eq. (107) that the bending moment at the built-in end depends on the position of the load *P*. If we equate to zero the derivative of eq. (107) with respect to *c*, we find that the moment *M_a* has its numerical maximum value when $c = l/\sqrt{3}$. Then

$$|M_a|_{\max} = \frac{Pl}{3\sqrt{3}} = 0.192Pl. \quad (108)$$

The bending moment under the load, from Fig. 155 (*d*), is

$$M_d = \frac{Pc(l-c)}{l} - \frac{c}{l} \frac{Pc(l^2 - c^2)}{2l^2} = \frac{Pc}{2l^3} (l-c)^2 (2l+c). \quad (b)$$

If we take the derivative of (*b*) with respect to *c* and equate it to zero, we find that *M_d* becomes a maximum when

$$c = \frac{l}{2} (\sqrt{3} - 1) = 0.366l.$$

Substituting this in eq. (*b*), we obtain

$$(M_d)_{\max} = 0.174Pl.$$

Comparing this with eq. (108), we find that in the case of a moving load the maximum normal stresses σ_x are at the built-in section.

Having the solution for the single concentrated load and

using the method of superposition, the problem can be solved for other types of transverse loading by simple extension of the above theory. Take, for instance, the case represented in Fig. 156. The moment at the support *A*, produced by any element *qdc* of the load, is obtained from eq. (107) by substituting *qdc* for *P*. The total moment *M_a* at the support will be

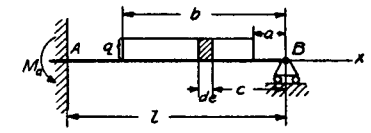


FIG. 156.

$$M_a = - \int_a^b \frac{qcdc(l^2 - c^2)}{2l^2} = - \frac{q}{2l^2} \left[\frac{l^2(b^2 - a^2)}{2} - \frac{b^4 - a^4}{4} \right]. \quad (c)$$

If the load be distributed along the entire length of the beam, then substituting in eq. (*c*) $a = 0$, $b = l$, we obtain

$$M_a = - \frac{ql^2}{8}. \quad (109)$$

The bending moment diagram is obtained by subtracting the triangular diagram due to the couple *M_a* (Fig. 157) from the parabolic diagram, due to uniform loading. It can

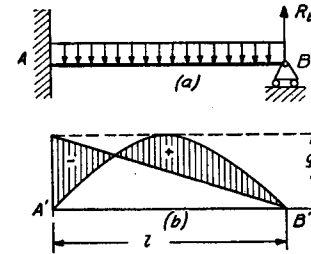


FIG. 157.

be seen that the maximum bending stresses will be at the built-in section. The deflection at any point is obtained by subtracting the deflection at this point produced by the couple *M_a* (see eq. 105, p. 159) from the deflection at the same point produced by the uniform load (see eq. 81, p. 138).

For the middle of the span we will obtain

$$\delta = \frac{5}{384} \frac{ql^4}{EI_z} + \frac{M_a l^2}{16EI_z} = \frac{ql^4}{192EI_z}. \quad (110)$$

Problems

1. Draw shearing force diagrams for the cases shown in Figures 155 and 157.

2. Determine the maximum deflection for the case of a uniformly distributed load shown in Fig. 157.

Solution. Combining eqs. (81) and (105), the following equation for the deflection curve is obtained:

$$y = \frac{q}{48EI_z} (3l^2x^2 - 5lx^3 + 2x^4). \quad (d)$$

Setting the derivative dy/dx equal to zero, we find the point of maximum deflection at $x = (l/16)(15 - \sqrt{33}) = 0.579l$. Substituting in (d), we obtain

$$\delta_{\max} = \frac{ql^4}{185EI_z}.$$

3. Determine the reaction at the right support of the beam shown in Fig. 157, considering this reaction as the redundant constraint.

Solution. Removing support B , the deflection of this end of the beam, considered as a cantilever, will be $ql^4/8EI_z$ from eq. (84). Reaction R_b at B (Fig. 157, *a*) must be such as to eliminate the above deflection. Then by using eq. (95) we obtain the equation:

$$\frac{ql^4}{8EI_z} = \frac{R_b l^3}{3EI_z},$$

from which

$$R_b = \frac{3}{8}ql.$$

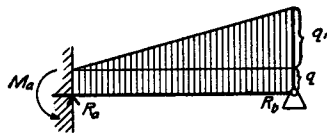


FIG. 158.

4. A beam is loaded as shown in Fig. 158. Determine the moment M_a and the reactions R_a and R_b at the supports.

Answer.

$$-M_a = \frac{ql^2}{8} + \frac{7}{120}ql^2; \quad R_a = \frac{5}{8}ql + \frac{9}{40}ql; \quad R_b = \frac{3}{8}ql + \frac{11}{40}ql.$$

5. Determine the reaction R_b at the support B of a uniformly loaded beam such as shown in Fig. 157 if the support B is elastic, so that a downward force of the magnitude k lowers the support a unit distance.

Solution. Using the same method as in problem 3 above, the

equation for determining R_b will be

$$\frac{ql^4}{8EI_z} - \frac{R_b l^3}{3EI_z} = \frac{R_b}{k},$$

from which

$$R_b = \frac{3}{8}ql \frac{1}{1 + \frac{3EI_z}{kl^3}}.$$

6. Construct the bending moment diagram for a uniformly loaded beam supported at three equidistant points.

Suggestion. From the condition of symmetry the middle cross section does not rotate during bending and each half of the beam will be in the condition of a beam built in at one end and supported at the other.

7. Determine the deflection of the end C of the beam shown in Fig. 159.

Solution. Replacing the action of the overhang by a couple Pa , the bending of the beam between the supports will be obtained by superposing cases (b) and (c). The statically indeterminate couple M_a will be found from the equation $\theta_1 = -\theta_1'$ or

$$\frac{Pal}{6EI_z} = \frac{M_a l}{3EI_z},$$

from which $M_a = Pa/2$. The deflection at C will be

$$\delta = \frac{Pa^3}{3EI_z} + a(\theta_2 - \theta_2') = \frac{Pa^3}{3EI_z} + \frac{Pa^2 l}{4EI_z}.$$

The first term on the right side represents the deflection of a cantilever and the second represents the deflection due to rotation of the cross section at B .

8. Determine the additional pressure of the beam AB on the support B (Fig. 155) due to non-uniform heating of the beam, provided that the temperature varies from t_0 at the bottom to t at the top of the beam according to a linear law.

Solution. If the support at B is removed, the non-uniform heating will cause the deflection curve of the beam to become an arc of a circle. The radius of this circle can be determined from the equation $1/r = \alpha(t - t_0)/h$, in which h = the depth of the beam

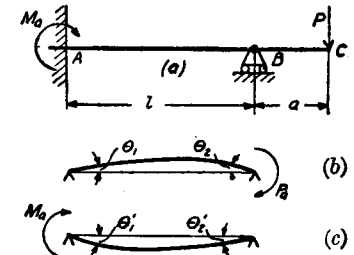


FIG. 159.

and α = the coefficient of thermal expansion. The corresponding deflection at B can be found as in problem 2, p. 94, and is

$$\delta = \frac{l^2}{2r} = \frac{l^2 \alpha (t - t_0)}{2h}.$$

This deflection is eliminated by the reaction of the support B . Letting R_b denote this reaction,

$$\frac{R_b l^3}{3EI_z} = \frac{l^2 \alpha (t - t_0)}{2h},$$

from which

$$R_b = \frac{3EI_z}{2hl} \cdot \alpha (t - t_0).$$

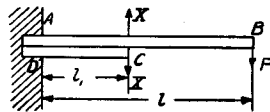


FIG. 160.

9. A cantilever AB , loaded at the end, is supported by a shorter cantilever CD of the same cross section as cantilever AB . Determine the pressure X between the two beams at C .

Solution. Pressure X will be found from the condition that at C both the cantilevers have the same deflection. Using eq. (95) for the lower cantilever and eq. (97) together with eq. (95) for the upper we obtain

$$\frac{Xl_1^3}{3EI_z} = \frac{P}{EI_z} \left(\frac{l_1^2}{2} - \frac{l_1^3}{6} \right) - \frac{Xl_1^3}{3EI_z},$$

from which

$$X = \frac{3P}{4} \left(\frac{l}{l_1} - \frac{1}{3} \right).$$

From a consideration of the bending moment diagrams for the upper and lower cantilevers it can be concluded that at C the upper cantilever has a larger angular deflection than the lower has. This indicates that there will be contact between the two cantilevers only at points D and C .

10. Solve problem 7 assuming, instead of a concentrated load P , a uniformly distributed load of intensity q to be distributed (1) along the length a of the overhand, (2) along the entire length of the beam.

11. Draw the bending moment and shearing force diagrams for the case shown in Fig. 156 if $a = 4$ ft., $b = 12$ ft., $l = 15$ ft. and $q = 400$ lbs. per ft.

42. Beam with Both Ends Built in.—In this case we have six reactive elements (three at each end), i.e., the problem has

three statically indeterminate elements. However, for ordinary beams, the horizontal components of the reactions can be neglected (see p. 178), which reduces the number of statically indeterminate quantities to two. Let us take the moments M_a and M_b at the supports for the statically indeterminate quantities. Then for the case of a single concentrated load P (Fig. 161, *a*) the solution can be obtained by combining the two statically determinate problems shown in Fig. 161 (*b*) and (*c*). It is evident that the conditions at the built-in ends of the beam AB will be satisfied if the couples M_a and M_b are adjusted so as to make

$$\theta_1 = -\theta_1'; \quad \theta_2 = -\theta_2'. \quad (a)$$

From these two equations the two statically indeterminate couples are obtained. Using eqs. (88) and (89) for a concentrated load and eqs. (103) and (104) for the couples, eqs. (*a*) become

$$\begin{aligned} -\frac{Pc(l^2 - c^2)}{6lEI_z} &= \frac{M_a l}{3EI_z} + \frac{M_b l}{6EI_z}, \\ -\frac{Pc(l - c)(2l - c)}{6lEI_z} &= \frac{M_a l}{6EI_z} + \frac{M_b l}{3EI_z}, \end{aligned}$$

from which

$$M_a = -\frac{Pc^2(l - c)}{l^2}; \quad M_b = -\frac{Pc(l - c)^2}{l^2}. \quad (111)$$

Combining the bending moment diagrams for cases (*b*) and (*c*), the diagram shown in Fig. 161 (*d*) is obtained. The maximum

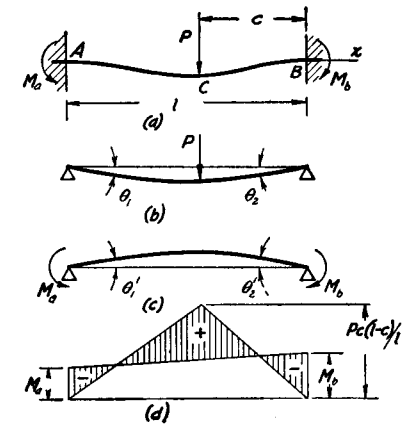


FIG. 161.

positive bending moment is under the load at the point C . Its magnitude can be found from Fig. 161 (d) and is given by the following:

$$M_c = \frac{Pc(l-c)}{l} + \frac{M_a c}{l} + \frac{M_b(l-c)}{l} = \frac{2Pc^2(l-c)^2}{l^3}. \quad (112)$$

From Fig. 161 (d) it may be seen that the numerically maximum bending moment is either that at C or that at the nearest support. For a moving load, i.e., when c varies, assuming $c < l/2$, the maximum numerical value of M_b is obtained by putting $c = 1/3l$ in eq. (111). This maximum is equal to $4/27Pl$. The bending moment under the load is a maximum when $c = l/2$ and this maximum, from eq. (112), is equal to $1/8Pl$. Hence for a moving load the greatest moment is at the end.

By using the method of superposition the deflection at any point can also be obtained by combining the deflection produced by load P with that produced by couples M_a and M_b .

Having the solution for a single concentrated load P , any other type of transverse loading can easily be studied by using the method of superposition.

Problems

1. Draw the shearing force diagram for the case in Fig. 161 (a) if $P = 1,000$ lbs., $l = 12$ ft., and $c = 4$ ft.

2. Construct the bending moment diagram for a uniformly loaded beam with built-in ends (Fig. 162).

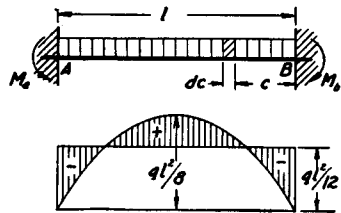


FIG. 162.

The moment produced by the load over the entire span is then

$$M_a = - \int_0^l \frac{q d c c^2 (l-c)}{l^2} = - \frac{q l^2}{12};$$

Solution. The moment at A produced by one element $q d c$ of the load (Fig. 162, a) is, from eq. (111),

$$dM_a = - \frac{q d c c^2 (l-c)}{l^2}.$$

the moment at the support B will have the same magnitude. Combining the parabolic bending moment diagram produced by the uniform load with the rectangular diagram given by two equal couples applied at the ends, we obtain the diagram shown in Fig. 162 (b) by the shaded area.

3. Determine the moments at the ends of a beam with built-in ends and loaded by the triangular load shown in Fig. 163.

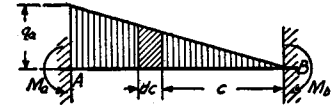


FIG. 163.

Solution. The intensity of the load at distance c from the support B is $q_a c / l$ and the load represented by the shaded element is $q_a c d c / l$. The couples acting at the ends, produced by this elementary load, as given by eqs. (111), are

$$dM_a = - \frac{q_a c^3 (l-c) d c}{l^3}; \quad dM_b = - \frac{q_a c^2 (l-c)^2 d c}{l^3};$$

therefore

$$M_a = - \int_0^l \frac{q_a c^3 (l-c) d c}{l^3} = - \frac{q_a l^2}{20};$$

$$M_b = - \int_0^l \frac{q_a c^2 (l-c)^2 d c}{l^3} = - \frac{q_a l^2}{30}.$$

4. Determine the reactive couples M_a and M_b in a beam with built-in ends and bent by a couple Pc (Fig. 164).

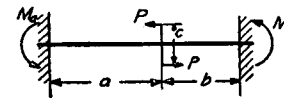


FIG. 164.

Solution. By using the solution of problem 5, p. 160, and eqs. (104) and (105), the following equations are obtained:

$$2M_a + M_b = - \frac{3Pc}{l^3} \left[a^2 \left(b + \frac{a}{3} \right) - \frac{2}{3} b^3 \right],$$

$$2M_b + M_a = \frac{3Pc}{l^3} \left[\frac{2}{3} a^3 - b^2 \left(a + \frac{b}{3} \right) \right],$$

from which M_a and M_b can easily be calculated.

5. Determine the bending moments at the ends of a built-in beam due to non-uniform heating of the beam if the temperature varies from t_0 at the bottom to t at the top of the beam according to a linear law.

Answer.

$$M_a = M_b = \frac{\alpha EI_z(t - t_0)}{h},$$

where α is the coefficient of thermal expansion and h is the depth of the beam.

6. Determine the effect on the reactive force and reactive couple at A of a small vertical displacement δ of the built-in end A of the beam AB (Fig. 161).

Solution. Remove the support A ; then the deflection δ_1 at A and the slope θ_1 at this point will be found as for a cantilever built in at B and loaded by P , i.e.,

$$\delta_1 = \frac{Pc^3}{3EI_z} + \frac{Pc^2}{2EI_z}(l - c); \quad \theta_1 = \frac{Pc^2}{2EI_z}.$$

Applying at A an upward reactive force X and a reactive couple Y in the same direction as M_a , of such magnitude as to annihilate the slope θ_1 and to make the deflection equal to δ , the equations for determining the unknown quantities X and Y become

$$\begin{aligned} \frac{Xl^2}{2EI_z} - \frac{Yl}{EI_z} &= \frac{Pc^2}{2EI_z}, \\ \frac{Xl^3}{3EI_z} - \frac{Yl^2}{2EI_z} &= \delta_1 - \delta. \end{aligned}$$

7. Draw the shearing force and bending moment diagrams for the beam shown in Fig. 163 if $q_a = 400$ lbs. per ft. and $l = 15$ ft.

8. Draw the shearing force and bending moment diagrams for a beam with built-in ends if the left half of the beam is uniformly loaded with a load $q = 400$ lbs. per ft. The length of the beam is $l = 16$ ft.

43. Frames.—The method used above for the cases of statically indeterminate beams can be applied also to the study of frames. Take, as a simple example, the symmetrical and symmetrically loaded frame, Fig. 165, hinged at C and D . The shape of the frame after deformation is shown by the dotted lines. Neglecting the change in the length of the bars and the effect of axial forces on the bending of bars,³ the frame can be considered to be made up of three beams as shown in

³ Simultaneous action of bending and thrust will be discussed later (see Part II).

Fig. 165 (b). It is evident that there will be couples M at the ends of the horizontal beam AB which oppose the free rotation of these ends and represent the action of the vertical bars on the horizontal beam. This couple M can be considered as the only statically indeterminate quantity. Knowing M , the

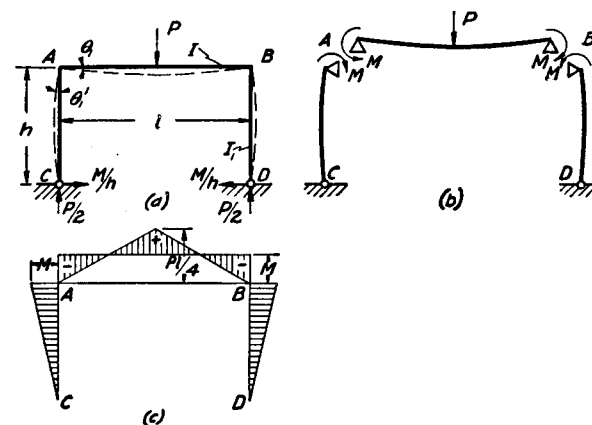


FIG. 165.

bending of all three bars can be investigated without any difficulty. For determining M we have the condition that at A and at B there are rigid joints between the bars so that the rotation of the top end of the vertical bar AC must be equal to the rotation of the left end of the horizontal bar. Hence the equation for determining M is

$$\theta_1 = \theta_1'. \quad (a)$$

θ_1 must be determined from the bending of the horizontal beam AB . Denoting by l the length of this beam and by EI its flexural rigidity, the rotation of the end A due to the load P , by eq. (88) ($b = l/2$), is $Pl^2/16EI$. The couples at the ends resist this bending and give a rotation in the opposite direction, which, from eqs. (103) and (104), equals $Ml/2EI$. The final value of the angle of rotation will be

$$\theta_1 = \frac{Pl^2}{16EI} - \frac{Ml}{2EI}.$$

Considering now the vertical bar AC as a beam with supported ends, bent by a couple M , and denoting by h its length and by EI_1 its flexural rigidity, the angle at the top, from eq. (104), will be

$$\theta_1' = \frac{Mh}{3EI_1}.$$

Substituting in eq. (a), we obtain

$$\frac{Pl^2}{16EI} - \frac{Ml}{2EI} = \frac{Mh}{3EI_1},$$

from which

$$M = \frac{Pl}{8} \frac{1}{1 + \frac{2h}{3l} \frac{I}{I_1}}. \quad (113)$$

This is the absolute value of M . Its direction is shown in Fig. 165 (b). Knowing M , the bending moment diagram can be constructed as shown in Fig. 165 (c). The reactive forces at the hinges C and D are also shown (Fig. 165, a). The vertical components of these forces, from considerations of symmetry, are each equal to $P/2$. As regards the horizontal components, their magnitude M/h is obtained by considering the vertical bars as simply supported beams loaded at the top by the couples M .

The same problem can be solved in another way by taking the horizontal reaction H at the hinges C and D as the statically indeterminate quantity, instead of M (Fig. 166). The statically indeterminate problem is solved by superposing the two statically determinate problems shown in Fig.

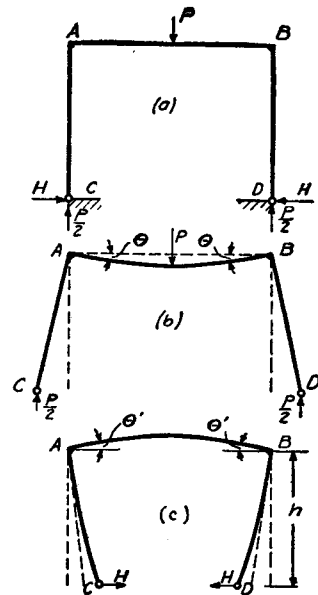


FIG. 166.

166 (b) and (c). In case (b) the redundant constraint prevent-

ing the horizontal motion of the hinges C and D is removed. The vertical bars no longer have any bending. The horizontal bar AB is in the condition of a bar with simply supported ends whose angles of rotation are equal to $Pl^2/16EI$, and the horizontal motion of each hinge C and D is therefore $h(Pl^2/16EI)$. In case (c) the effect of the forces H is studied. These forces produce bending couples on the ends of the horizontal bar AB equal to $H \cdot h$, so that the angles of rotation of its ends θ' will be $Hh \cdot l/2EI$. The deflection of each hinge C and D consists of two parts, the deflection $\theta' h = Hh^2 l/2EI$ due to rotation of the upper end and the deflection $Hh^3/3EI_1$ of the vertical bars as cantilevers. In the actual case (Fig. 166, a) the hinges C and D do not move; hence the horizontal displacements produced by the force P (Fig. 166, b) must be counteracted by the forces H (Fig. 166, c), i.e.,

$$\frac{Pl^2}{16EI} h = \frac{Hh^2 l}{2EI} + \frac{Hh^3}{3EI_1},$$

from which

$$H = \frac{1}{h} \frac{Pl}{8} \frac{1}{1 + \frac{2h}{3l} \cdot \frac{I}{I_1}}.$$

Remembering that $Hh = M$, this result agrees with the equation (113) above.

This latter method of analysis is especially useful for nonsymmetrical loading such as shown in Fig. 167. Removing the constraint preventing the hinges C and D from horizontal motion, we have the condition represented in Fig. 167 (b). It is evident that the increase in distance between C and D may be obtained by multiplying by h the sum of the angles θ_1 and θ_2 . Using eqs. (88) and (89), this increase in distance

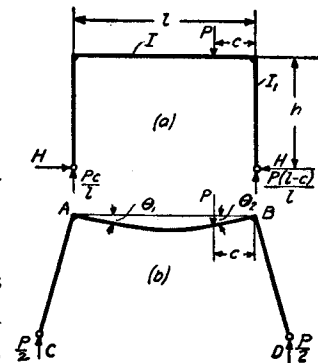


FIG. 167.

becomes

$$h \left[\frac{Pc(l^2 - c^2)}{6EI} + \frac{Pc(l - c)(2l - c)}{6EI} \right] = \frac{Pc(l - c)h}{2EI}.$$

It must be eliminated by the horizontal reactions H (Fig. 166, c). Then, using the results obtained in the previous problem, we get the following equation for determining H :

$$2 \left(\frac{Hh^2l}{2EI} + \frac{Hh^3}{3EI_1} \right) = \frac{Pc(l - c)h}{2EI},$$

from which

$$H = \frac{Pc(l - c)}{2hl} \frac{I}{1 + \frac{2}{3} \frac{I}{I_1} \frac{h}{l}}. \quad (114)$$

Having the solution for one concentrated load, any other case of loading of the beam AB of the frame can easily be studied by the method of superposition.

Let us consider now a frame with built-in supports and an unsymmetrical loading as shown in Fig. 168. In this case we have

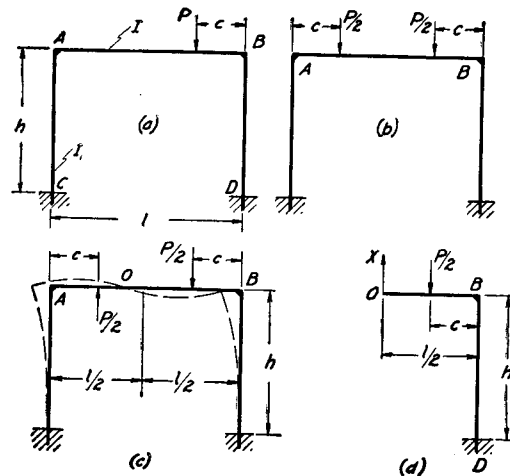


FIG. 168.

three reactive elements at each support and the system has three statically indeterminate elements. In the solution of this problem

we will use a method based on the method of superposition in which the given system of loading is split into parts such that for each partial loading a simple solution can be found.⁴ The problem shown in Fig. 168 (a) can be solved by superposing the solutions of the two problems shown in Fig. 168 (b) and (c). The case shown in (b) is a symmetrical one and can be considered in the same manner as the first example shown in Fig. 165. A study of the case shown in (c) will show that the point of inflection O of the horizontal bar AB is situated at the middle of the bar. This follows from the condition that the loads $P/2$ are equally distant from the vertical axis of symmetry of the frame and are opposite in sense. The moment, the deflection, and the axial force produced at the mid point O of the horizontal beam AB by one of the loads $P/2$ will be removed by the action of the other load $P/2$. Hence there will be no bending moment, no vertical deflection and no axial force at O . The magnitude of the shearing force at the same point X can be found from the condition that the vertical deflection of O is equal to zero (Fig. 168, d). This deflection consists of two parts, a deflection δ_1 due to the bending of the cantilever OB and a deflection δ_2 due to rotation of the end B of the vertical bar BD . Using the known equations for a cantilever (eq. 98), and using the notations given in the figure, the following equations are obtained:

$$\delta_1 = \frac{P}{2} \frac{c^3}{3EI} + \frac{P}{2} \frac{c^2}{2EI} \left(\frac{l}{2} - c \right) - \frac{X \left(\frac{l}{2} \right)^3}{3EI},$$

$$\delta_2 = \left(\frac{Pc}{2} - X \frac{l}{2} \right) \frac{h}{EI_1} \frac{l}{2}.$$

Substituting this in the equation $\delta_1 + \delta_2 = 0$, the magnitude X of the shearing force can be found. Having determined X , the bending moment at every cross section of the frame for case (c) can be calculated. Combining this with the bending moments for the symmetrical case (b), the solution of the problem (a) is obtained.⁵

Problems

1. Determine the bending moments at the corners of the frame shown in Fig. 169.

⁴ Such a method was extensively used by W. L. Andréé; see his book "Das B-U Verfahren," München and Berlin, 1919.

⁵ Solutions of many important problems on frames can be found in the book by Kleinlogel, "Mehrstielige Rahmen," Berlin, 1927.

Solution. Considering the bar AB as a beam supported at the

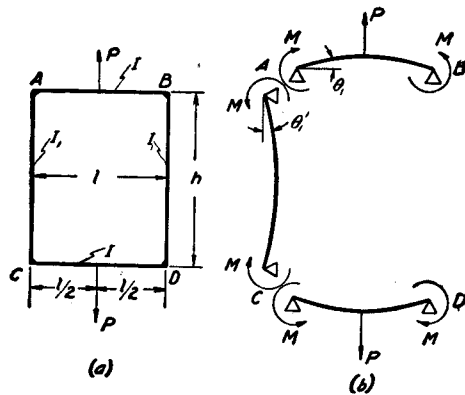


FIG. 169.

ends (Fig. 169, b) and denoting by M the moments at the corners, the angle θ_1 will be

$$\frac{Pl^2}{16EI} - \frac{Ml}{2EI}.$$

Putting this equal to the angle θ_1' at the ends of the vertical bars which are bent by the couples M only, the following equation for M is obtained:

$$\frac{Pl^2}{16EI} - \frac{Ml}{2EI} = \frac{Mh}{2EI_1},$$

from which

$$M = \frac{Pl}{8} \frac{1}{1 + \frac{h}{l} \frac{I}{I_1}}.$$

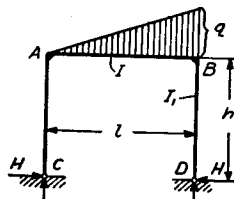


FIG. 170.

2. Determine the horizontal reactions H for the case shown in Fig. 170.

Suggestion. By using eq. (114) and applying the method of superposition, we get

$$H = \frac{ql^2}{24h} \frac{1}{1 + \frac{2}{3} \frac{I}{I_1} \frac{h}{l}}.$$

3. Draw the bending moment diagram for the three bars of the preceding problem assuming $h = l$ and $I = I_1$.

4. Determine the bending moments at the joints of the frame shown in Fig. 171.

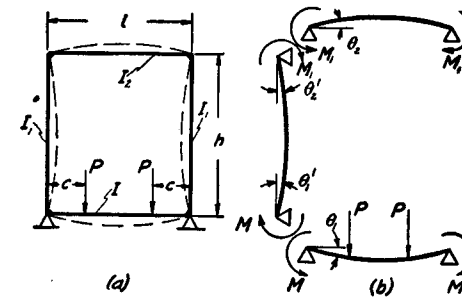


FIG. 171.

Solution. Disjoining the frame as shown in Fig. 171 (b), the equations for determining the couples M and M_1 are

$$\theta_1 = \theta_1' \quad \text{and} \quad \theta_2 = \theta_2'.$$

Substituting in these equations

$$\theta_1 = \frac{Pc(l-c)}{2EI} - \frac{Ml}{2EI};$$

$$\theta_1' = \frac{Mh}{3EI_1} - \frac{M_1h}{6EI_1};$$

$$\theta_2 = \frac{M_1l}{2EI_2};$$

$$\theta_2' = \frac{Mh}{6EI_1} - \frac{M_1h}{3EI_1},$$

we obtain two equations for determining M and M_1 .

5. A symmetrical rectangular frame is submitted to the action of a horizontal force H as shown in Fig. 172. Determine the bending moments M and M_1 at the joints.

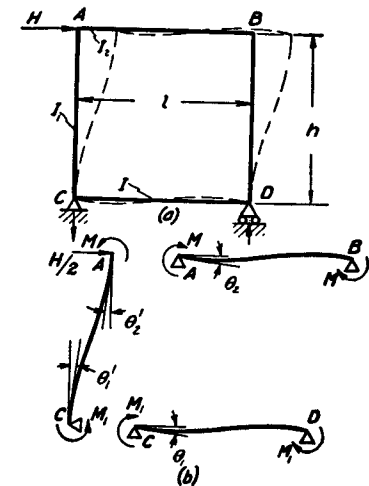


FIG. 172.

Solution. The deformed shape of the frame is shown in Fig. 172 (a). Disjoining the frame as shown in Fig. 172 (b), and applying moments the directions of which are chosen to comply with the distorted shape of the frame, Fig. 172 (a), we have

for the bar CD

$$\theta_1 = \frac{M_1 l}{6EI} = \left(\frac{Hh}{2} - M \right) \frac{l}{6EI}. \quad (a)$$

Considering now the vertical bar AC as a cantilever built in at the end C at an angle θ_1 , the slope at the end A will be

$$\theta_2' = \theta_1 + \frac{H}{2} \frac{h^2}{2EI_1} - \frac{Mh}{EI_1}. \quad (b)$$

Finally, due to bending of the bar AB ,

$$\theta_2 = \theta_2' = \frac{Ml}{6EI_2}. \quad (c)$$

Then, from eqs. (a), (b) and (c),

$$M = \frac{Hh}{2} \left(1 + \frac{3h}{l} \frac{I}{I_1} \right) \frac{1}{1 + \frac{I}{I_2} + 6 \frac{h}{l} \frac{I}{I_1}}. \quad (d)$$

Substituting in eq. (a), the bending moment M_1 can be found. When the horizontal bar CD has very great rigidity, we approach the condition of the frame shown in Fig. 168, submitted to a lateral load H . Substituting in (d) $I = \infty$, we obtain for this case

$$M = \frac{Hh}{4} \frac{1}{1 + \frac{1}{6} \frac{l}{h} \frac{I_1}{I_2}}. \quad (e)$$

The case of a frame such as shown in Fig. 165 with hinged supports and submitted to the action of a lateral load applied at A can also be obtained by substituting $I = 0$ in eq. (d).

6. Determine the horizontal reactions H , and the moments M_a and M_b at the joints A and B for the frame shown in Fig. 173.

Answer.

$$H = \frac{qh}{20} \frac{11m + 20}{2m + 3},$$

$$M_a = M_b = \frac{qh^2}{60} \frac{7m}{2m + 3},$$

where

$$m = \frac{I}{I_1} \cdot \frac{h}{l}.$$

7. A frame consists of two bars joined rigidly at B and built in at A and C (Fig. 174). Determine the bending moment M at

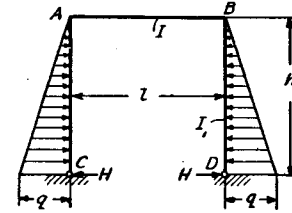


FIG. 173.

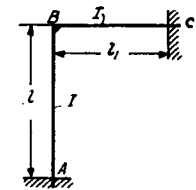


FIG. 174.

B and the compressive force P in AB when, due to a rise in temperature, the bar AB increases in length by $\Delta = \alpha l(t - t_0)$.

Answer. P and M can be found from equations:

$$\frac{Pl_1^3}{3EI} - \frac{Ml_1^2}{2EI} = \Delta,$$

$$\frac{Pl_1^2}{2EI} - \frac{Ml_1}{EI} = \frac{Ml}{4EI}.$$

44. Beams on Three Supports.—In the case of a beam on three supports (Fig. 175, *a*) there is one statically indeterminate reactive element. Let the reaction of the intermediate support be this element. Then by using the method of superposition the solution of case (*a*) may be obtained by combining the cases represented in (*b*) and (*c*), Fig. 175. The intermediate reaction X is found by using the condition that the deflection δ produced at C by the load P must be eliminated by the reaction X . Using eq. (86), we get the following equation for determining X :

$$\frac{Pcl_1[(l_1 + l_2)^2 - c^2 - l_1^2]}{6(l_1 + l_2)EI_z} = \frac{Xl_1^2l_2^2}{3(l_1 + l_2)EI_z},$$

from which

$$X = \frac{Pc[(l_1 + l_2)^2 - c^2 - l_1^2]}{2l_1l_2^2}. \quad (115)$$

If P is acting on the left span of the beam, the same equation can be used, but the distance c must be measured from the

support A and l_1 and l_2 must be interchanged. For $l_1 = l_2 = l$, from (115),

$$X = \frac{Pc(3l^2 - c^2)}{2l^3}. \quad (116)$$

Having the solution for a single load P , any other loading can easily be studied by using the method of superposition.

The same problem can be solved in another manner. Imagine the beam cut into two parts at C (Fig. 175, d) and let

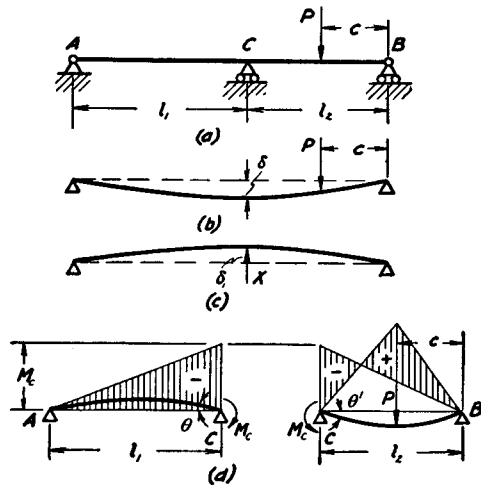


FIG. 175.

M_c denote the magnitude of the bending moment of the original beam at this cross section. In this manner the problem is reduced to the consideration of the two simply supported beams shown in (d) which are statically determinate. The magnitude of M_c is determined from the condition of continuity of the deflection curve at the support C . From this $\theta = \theta'$, whence, using the eqs. (88), p. 142, and (104), p. 158, we obtain

$$-\frac{M_c l_1}{3EI_z} = \frac{Pc(l_2^2 - c^2)}{6l_2 EI_z} + \frac{M_c l_2}{3EI_z},$$

from which

$$M_c = -\frac{Pc(l_2^2 - c^2)}{2l_2(l_1 + l_2)}. \quad (117)$$

The direction of M_c as determined from the condition of bending is shown by arrows in Fig. 175 (d). The bending moment diagram is shown by the shaded area in Fig. 175 (d).

Problems

1. For the example in Fig. 175 prove that the magnitude of the bending moment M_c given by eq. (117) is the same as that given for the cross section C by eq. (115).

2. Draw the shearing force diagram for the beam of the preceding problem if $l_1 = l_2$, $c = l_2/2$, and $P = 1,000$ lbs.

3. A beam on three supports (Fig. 175, a) is carrying a uniformly distributed load of intensity q . Determine the bending moment at the support C .

Solution. By the method of superposition, substituting qdc for P in eq. (117) and integrating along both spans, we obtain

$$M_c = -\int_0^{l_2} \frac{qc(l_2^2 - c^2)dc}{2l_2(l_1 + l_2)} - \int_0^{l_1} \frac{qc(l_1^2 - c^2)dc}{2l_1(l_1 + l_2)} = -\frac{q}{8} \frac{l_2^3 + l_1^3}{l_1 + l_2},$$

when

$$l_1 = l_2 = l, \quad M_c = -\frac{ql^2}{8}.$$

The direction of this moment is as shown in Fig. 175 (d).

4. Draw the shearing force diagram for the preceding problem assuming $l_1 = l_2$ and $q = 500$ lbs. per ft.

5. Determine the numerical maximum bending moment in the beam ACB (Fig. 175) if $P = 10,000$ lbs., $l_1 = 9$ ft., $l_2 = 12$ ft., $c = 6$ ft.

Answer. $M_{\max} = 23,600$ lbs. ft.

6. A beam on three equidistant supports is carrying a uniformly distributed load of intensity q . What effect will it have on the middle reaction if the middle support is lowered a distance δ ?

Solution. Using the method shown in Fig. 175 (b) and (c), the middle reaction X is found from the equation:

$$\frac{5}{384} \frac{q(2l)^4}{EI} = \frac{X(2l)^3}{48EI} + \delta,$$

from which

$$X = \frac{5}{8} 2ql - \frac{6\delta EI}{l^3}.$$

7. Determine the additional pressure of the beam AB on the support C (Fig. 175, a) due to non-uniform heating of the beam

if the temperature varies from t at the bottom to t_1 at the top of the beam according to a linear law, $t > t_1$ and $l_1 = l_2 = l$.

Solution. If the support at C were removed, then, due to the non-uniform heating, the deflection curve of the beam would become the arc of a circle. The radius of this circle is determined by the equation:

$$\frac{1}{r} = \frac{\alpha(t - t_1)}{h},$$

in which h = the depth of the beam and α = the coefficient of thermal expansion. The corresponding deflection at the middle is $\delta = l^2/2r$ and the reaction X at C can be found from the equation:

$$\frac{X(2l)^3}{48EI} = \delta.$$

8. Determine the bending moment diagram for the beam ABC supported by three pontoons (Fig. 176) if the horizontal cross-sectional area of each pontoon is A and the weight of unit volume of water is γ .

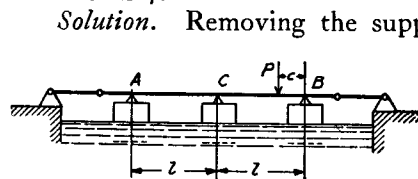


FIG. 176.

Solution. Removing the support at C , the deflection δ produced at this point by the load P consists of two parts: (a) the deflection due to bending of the beam and (b) the deflection due to sinking of pontoons A and B . From eq. (91) we obtain

$$\delta = \frac{Pc}{48EI_z} [3(2l)^2 - 4c^2] + \frac{P}{2A\gamma}. \quad (a)$$

The reaction X of the middle support diminishes the above deflection by

$$\frac{X(2l)^3}{48EI_z} + \frac{X}{2A\gamma}. \quad (b)$$

The difference between (a) and (b) represents the distance the middle pontoon sinks, from which we get the following equation for determining X :

$$\frac{Pc}{48EI_z} [3(2l)^2 - 4c^2] + \frac{P}{2A\gamma} - \frac{X(2l)^3}{48EI_z} - \frac{X}{2A\gamma} = \frac{X}{A\gamma}.$$

Knowing X , the bending moment diagram can readily be obtained.

45. Continuous Beams.—In the case of a continuous beam on many supports (Fig. 177) one support is usually considered as an immovable hinge while the other supports are hinges on rollers. In this arrangement every intermediate support has only one unknown reactive element, the magnitude of the vertical reaction; hence the number of statically indeterminate

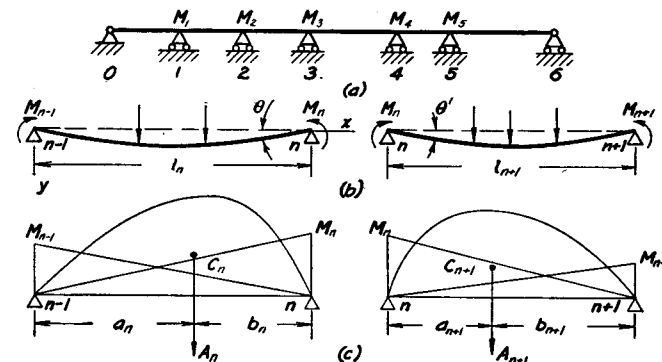


FIG. 177.

elements is equal to the number of intermediate supports. For instance, in the case shown in Fig. 177 (a) the number of statically indeterminate elements is five. Both methods shown in the previous article can be used here also. But if the number of supports is large, the second method, in which the bending moments at the supports are taken as the statically indeterminate elements, is by far the simpler method. Let Fig. 177 (b) represent two adjacent spans n and $n + 1$ of a continuous beam cut at supports $n - 1$, n and $n + 1$. Let M_{n-1} , M_n and M_{n+1} denote the bending moments at these supports. The directions of these moments depend on the loads on the beam. We will assume the directions shown in the figure.⁶ It is evident that if the bending moments at the supports are known the problem of the continuous beam will be reduced to that of calculating as many simply supported

⁶ If finally we would obtain negative signs for some moments, this will indicate that the directions of these moments are opposite to that shown in the figure.

beams as there are spans in the continuous beam. For calculating the bending moments M_{n-1} , M_n , M_{n+1} the condition of continuity of the deflection curve at the supports will be used. For any support n this condition of continuity is satisfied if the deflection curves of the two adjacent spans have a common tangent at the support n , i.e., if the slope at the right end of span n is equal to the slope at the left end of span $n+1$. To calculate these slopes the area moment method will be used. Let A_n denote the area of the bending moment diagram for the span n , considered as a simply supported beam, due to the actual load on this span; let a_n and b_n represent the horizontal distances of the centroid C_n of the moment area from the supports $n-1$ and n . Then the slope at the right end for this condition of loading is (see Art. 35)

$$-\frac{A_n a_n}{l_n EI_z}.$$

In addition to the deflection caused by the load on the span itself, the span n is also bent by the couples M_{n-1} and M_n . From equations (103) and (104) the slope produced at the support n by these couples is

$$-\left(\frac{M_n l_n}{3EI_z} + \frac{M_{n-1} l_n}{6EI_z}\right).$$

The total angle of rotation is then *

$$\theta = -\left(\frac{M_n l_n}{3EI_z} + \frac{M_{n-1} l_n}{6EI_z} + \frac{A_n a_n}{l_n EI_z}\right). \quad (a)$$

In the same manner, for the left end of the span $n+1$, we obtain

$$\theta' = \frac{A_{n+1} b_{n+1}}{l_{n+1} EI_z} + \frac{M_n l_{n+1}}{3EI_z} + \frac{M_{n+1} l_{n+1}}{6EI_z}. \quad (b)$$

From the condition of continuity it follows that

$$\theta = \theta'. \quad (c)$$

* The angle is taken positive if rotation is in clockwise direction.

Substituting expressions (a) and (b) in this equation we obtain

$$M_{n-1} l_n + 2M_n(l_n + l_{n+1}) + M_{n+1} l_{n+1} = -\frac{6A_n a_n}{l_n} - \frac{6A_{n+1} b_{n+1}}{l_{n+1}}. \quad (118)$$

This is the well-known *equation of three moments*.⁷ It is evident that the number of these equations is equal to the number of intermediate supports and the bending moments at the supports can be calculated without difficulty.

In the beginning it was assumed that the ends of the continuous beam were supported. If one or both ends are built in, then the number of statically indeterminate quantities will be larger than the number of intermediate supports and derivation of additional equations will be necessary to express the condition that no rotation occurs at the built-in ends (see problem 5 below).

Knowing the moments at the supports, there is no difficulty in calculating the reactions at the supports of a continuous beam. Taking, for instance, the two adjacent spans n and $n+1$ (Fig. 177, *b*), and considering them as two simply supported beams, the reaction R_n' at the support n , due to the loads on these two spans, can easily be calculated. In addition to this there will be a reaction due to the moments M_{n-1} , M_n and M_{n+1} . Taking the directions of these moments as indicated in Fig. 177 (*b*), the additional pressure on the support n will be

$$\frac{M_{n-1} - M_n}{l_n} + \frac{-M_n + M_{n+1}}{l_{n+1}}.$$

Adding this to the above reaction R_n' , the total reaction will be

$$R_n = R_n' + \frac{M_{n-1} - M_n}{l_n} + \frac{-M_n + M_{n+1}}{l_{n+1}}. \quad (119)$$

If concentrated forces are applied at the supports they will

⁷ This equation was established by Bertot; see *Comptes rendus de la Société des Ingenieurs civils*, p. 278, 1855; see also Clapeyron, *Paris, C. R.*, t. 45 (1857).

be transmitted directly to the corresponding supports and must be added to the right side of equation (119).

The general equation of continuity (*c*) can also be used for those cases where, by mis-alignment, the supports are not situated on the same level (Fig. 178). Let β_n and β_{n+1} denote the angles of inclination to the horizontal of the straight lines connecting the points of supports in the n th and $(n+1)$ th spans. The angle of rotation given by eqs. (a) and (b) was measured from the line connecting the centers of the hinges; hence the angle θ between the tangent at n and the horizontal line will be, for the span n ,

$$\theta = - \left(\frac{M_n l_n}{3EI_z} + \frac{M_{n+1} l_n}{6EI_z} + \frac{A_n a_n}{l_n EI_z} - \beta_n \right).$$

In the same manner for the span $n+1$

$$\theta' = \frac{A_{n+1} b_{n+1}}{l_{n+1} EI_z} + \frac{M_n l_{n+1}}{3EI_z} + \frac{M_{n+1} l_{n+1}}{6EI_z} + \beta_{n+1}.$$

Equating these angles we obtain

$$M_{n-1} l_n + 2M_n(l_n + l_{n+1}) + M_{n+1} l_{n+1} = - \frac{6A_n a_n}{l_n} - \frac{6A_{n+1} b_{n+1}}{l_{n+1}} - 6EI_z(\beta_{n+1} - \beta_n). \quad (120)$$

If h_{n-1} , h_n , h_{n+1} denote the vertical heights of the supports $n-1$, n and $n+1$ above a horizontal reference line, we have

$$\beta_n = \frac{h_{n-1} - h_n}{l_n}; \quad \beta_{n+1} = \frac{h_n - h_{n+1}}{l_{n+1}}.$$

Substituting in eq. (120), the bending moments at the supports due to mis-alignment can easily be calculated.

Problems

1. Determine the bending moment and shearing force diagrams for a continuous beam with three equal spans carrying a uniformly distributed load of intensity q (Fig. 179).

Solution. For a simply supported beam and a uniformly distributed load the bending moment diagram is a parabola with

maximum ordinate $ql_n^2/8$. The area of the parabolic segment is

$$A_n = \frac{2}{3} l_n \frac{ql_n^2}{8} = \frac{ql_n^3}{12}.$$

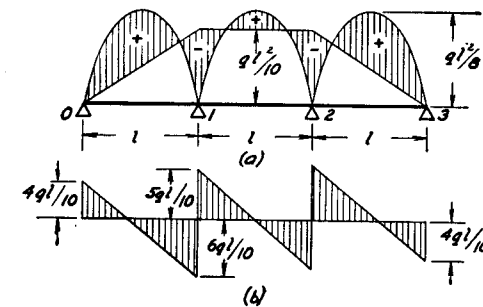


FIG. 179.

The centroid is at the middle of the span, so $a_n = b_n = l_n/2$. Substituting in eq. (118), we obtain

$$M_{n-1} l_n + 2M_n(l_n + l_{n+1}) + M_{n+1} l_{n+1} = - \frac{ql_n^3}{4} - \frac{q(l_{n+1})^3}{4}. \quad (118')$$

Applying this equation to our case (Fig. 179) for the first and the second span and noting that at the support 0 the bending moment is zero, we obtain

$$4M_1 l + M_2 l = - \frac{ql^3}{2}. \quad (d)$$

From the condition of symmetry it is evident that $M_1 = M_2$. Then, from (d), $M_1 = -(ql^2/10)$. The bending moment diagram is shown in Fig. 179 (a) by the shaded area. The reaction at support 0 is

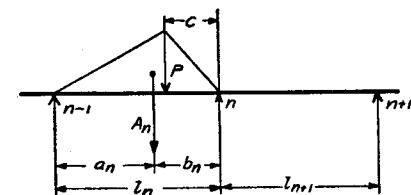


FIG. 180.

$$R_0 = \frac{ql}{2} - \frac{ql^2}{10l} = \frac{4}{10} ql.$$

The reaction at support 1 is

$$R_1 = ql + \frac{ql^2}{10l} = \frac{11}{10} ql.$$

The shearing force diagram is shown in Fig. 179 (b). The maximum moment will evidently be at a distance $4l/10$ from the ends of the

beam, where the shearing force is zero. The numerical maximum bending moment is at the intermediate supports.

2. Set up the expression for the right side of eq. (118) when there is a concentrated force in the span n and no load in the span $n + 1$ (Fig. 180).

Solution. In this case A_n is the area of the triangle of height $Pc(l_n - c)/l_n$ and with the base l_n ; hence $A_n = Pc(l_n - c)/2$ and $a_n = l_n - b_n = l_n - (l_n + c)/3$. Substituting in (118), we get

$$M_{n-1}l_n + 2M_n(l_n + l_{n+1}) + M_{n+1}l_{n+1} = -\frac{Pc(l_n - c)(2l_n - c)}{l_n}.$$

3. Determine the bending moments at the supports and the reactions for the continuous beam shown in Fig. 181.

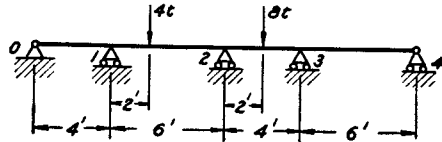


FIG. 181.

Answer. $M_1 = -1.54$ ton ft.; $M_2 = -3.74$ ton ft.; $M_3 = -1.65$ ton ft. The reactions are $R_0 = -0.386$ ton; $R_1 = 2.69$ tons; $R_2 = 6.22$ tons; $R_3 = 3.75$ tons; $R_4 = -0.275$ ton. The moments at the supports are negative and produce bending convex up.

4. Construct the bending moment and shearing force diagrams for the continuous beam shown in Fig. 182 (a) if $P = ql$, $c = l/4$.

Solution. In this case the imaginary loading for the first span is $A_1 = ql^3/12$, for the second span $A_2 = 0$, and for the third span

$$A_3 = \frac{Pc(l - c)}{2}; \quad a_3 = \frac{2l - c}{3}; \quad b_3 = \frac{l + c}{3}.$$

Substituting in eq. (118) gives the following equations for determining the bending moments M_1 and M_2 :

$$4M_1l + M_2l = -\frac{ql^3}{4},$$

$$M_1l + 4M_2l = -\frac{Pc(l^2 - c^2)}{l},$$

from which

$$M_1 = -\frac{49}{960}ql^2; \quad M_2 = -\frac{44}{960}ql^2.$$

Both these moments are negative, so that the bending moment diagram will be as shown in Fig. 182 (b). To obtain the shearing

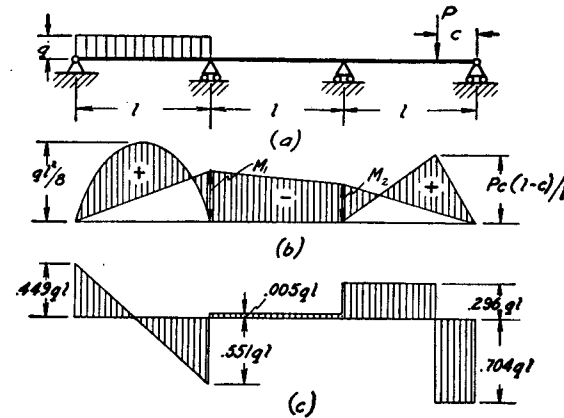


FIG. 182.

force diagram it is necessary to find the reactions at the supports of the separate spans (eq. 119). The pressures on supports 0 and 1 for the first span of the beam are

$$\frac{ql}{2} + \frac{M_1}{l} = 0.449ql \quad \text{and} \quad \frac{ql}{2} - \frac{M_1}{l} = 0.551ql.$$

The pressure on supports 1 and 2 of the second span of the beam are

$$\frac{-M_1 + M_2}{l} = 0.005ql \quad \text{and} \quad \frac{-M_2 + M_1}{l} = -0.005ql$$

and on supports 2 and 3 of the third span

$$\frac{Pc}{l} - \frac{M_2}{l} = 0.296ql \quad \text{and} \quad \frac{P(l - c)}{l} + \frac{M_2}{l} = 0.704ql.$$

From this, the shearing force diagram is obtained as shown in Fig. 182 (c).

5. Determine the bending moment diagram for the case shown in Fig. 183 (a).

Solution. Equation (118) for this case becomes

$$M_0l + 4M_1l + M_2l = 0.$$

Now, $M_2 = -Pc$, while the condition at the built-in end (sup-

port o) gives (from eqs. 103, 104)

$$\frac{M_0 l}{3EI} + \frac{M_1 l}{6EI} = 0.$$

From the above equations we obtain $M_0 = -\frac{1}{7}Pc$; $M_1 = +\frac{2}{7}Pc$; $M_2 = -Pc$. The bending moment diagram is as shown in Fig. 183 (b).

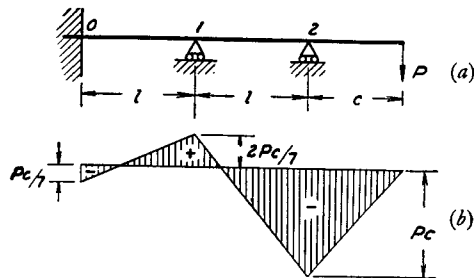


FIG. 183.

6. Determine the bending moments at the supports of a continuous beam with seven equal spans when the middle span alone is loaded by a uniformly distributed load q .

Answer.

$$M_3 = M_4 = -\frac{ql^2}{18.9}; \quad M_2 = M_5 = -\frac{1}{3.75}M_3; \quad M_1 = M_6 = \frac{1}{15}M_3.$$

7. A continuous beam having four equal spans of length 16 ft. each is uniformly loaded over the last span. Draw the shearing force and bending moment diagrams if $q = 400$ lbs. per ft.

8. Solve problem 5 assuming that a uniform load of intensity q is distributed along the entire length of the beam and that $c = l/2$. Draw the shearing force diagram for this loading condition.

CHAPTER VII

BEAMS OF VARIABLE CROSS SECTION. BEAMS OF TWO MATERIALS

46. Beams of Variable Cross Section.—In the preceding discussion all of the beams considered were of prismatical form. More elaborate investigation shows that equations (56) and (57), which were derived for prismatical bars, can also be used with sufficient accuracy for bars of variable cross section provided the variation is not too extreme. Cases of abrupt changes in cross section, in which considerable stress concentration takes place, will be discussed in Part II.

As a first example of a beam of a variable cross section let us consider the deflection of a cantilever beam of *uniform strength*, i.e., a beam in which the section modulus varies along the beam in the same proportion as the bending moment. Then, as is seen from eqs. (60), $(\sigma_x)_{\max}$ remains constant along the beam and can be taken equal to σ_w . Such a condition is favorable as regards the amount of material used, because each cross section will have only the area necessary to satisfy the conditions of strength.

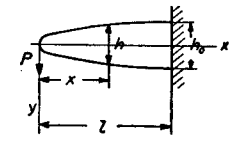


FIG. 184.

For a cantilever with an end load (Fig. 184), the bending moment at a cross section a distance x from the load is numerically equal to Px . In order to have a beam of uniform strength the section modulus also must be proportional to x . This condition can be fulfilled in various ways.

Let us take as a first example the case of a rectangular cross section of constant width b and variable depth h . From the definition of the beam of equal strength it follows that

$$\frac{M}{Z} = \frac{6Px}{bh^2} = \frac{6Pl}{bh_0^2} = \text{const.},$$